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單色 k 子棋

Monochromatic k in a row

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本論文係古國翰（R11221008）在國立臺灣大學數學系完成之碩士學位論文，
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摘要

本篇論文將探討一種 k 子棋的變體（即單色 k 子棋）：兩位玩家輪流在給定的棋盤上下棋，其中雙方玩家所使用的棋子沒有顏色差異。遊戲會在棋盤上產生了一條 k 子連線時結束，由下最後一子的玩家獲勝。和一般的 k 子棋不同，玩家不再需要佔領一整條 k 子連線；僅需佔領某條 k 子連線的最後一個空位即可。

本篇聚焦在此類遊戲的總步數估計。嚴格來說，是對最終棋盤上棋子（之於棋盤格數）的密度做估計。作爲原始遊戲棋盤的延伸，平面網格（$\mathbb{Z}^2$）和超立方體（$[k]^d$）是本篇主要探討的兩種棋盤類型。

就最小密度而言，對於平面網格棋盤上的單色 3 子棋，我們有確切值：$\frac{1}{17}$。對於其他更大的 k，其介在 $1-\frac{16}{k}-o(k^{-1})$ 和 $1-\frac{8}{k}+o(k^{-1})$ 之間。對於超立方體棋盤，最小密度大致爲 $1-\frac{2d}{k}+\frac{2d(d-1)}{k^2}\pm O(k^{-3})$。

就最大密度而言，我們仍有平面棋盤上單色 3 子棋的確切值：$\frac{1}{5}$。對於不是 3 的倍數的 k，其等於 $1-\frac{2}{k}$。對於剩餘的 k，上下界非常相近，其值介於 $1-\frac{2}{k-1}$ 和 $1-\frac{2}{k}$ 之間。超立方體的則是 $1-\frac{2}{k}-O(k^{-2})$。

關鍵字：k 子棋、五子棋、位置遊戲、極值問題、Nikodym 集合問題

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Abstract

We introduce a variant of k in a row: On a given board, each of players claims a position in turns until there is a k in a row among **all** claimed positions. The difference to the original k in a row is that, in this variant, a player no longer needs to claim a **full** k in a row by themself to win the game; instead, claiming **the last remaining position** of a k in a row is sufficient.

We are curious about how long a game can last, or, equivalently, how dense a final board configuration can be. Likewise, we concern how sparse a configuration can be. We ask the same questions for various boards, say the infinite plane board and the hypercubes $[k]^d$ for every $d \in \mathbb{N}$. To answer it, we give bounds on the maximum and minimum densities configurations.

Speaking minimum density, on the infinite plane board, we have the exact answer, $\frac{1}{17}$, when $k = 3$. For other k, the minimum density is between $1 - \frac{16}{k} - o(k^{-1})$ and $1 - \frac{8}{k} + o(k^{-1})$. On the hypercube $[k]^d$, it is around $1 - \frac{2d}{k} + \frac{2d(d-1)}{k^2} \pm O(k^{-3})$.

Speaking maximum density, we still have the exact answer for the plane board when $k = 3$, which is $\frac{1}{5}$. For other k with $3 \nmid k$, it is exactly $1 - \frac{2}{k}$. For other multiple of 3, it is between $1 - \frac{2}{k-1}$ and $1 - \frac{2}{k}$. On the hypercube, it is around $1 - \frac{2}{k} - O(k^{-2})$.

Keywords: k in a row, Five in a row, Gomoku, Positional Game, Extremal Problem, Nikodym set Problem

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	Infinite Plane $\mathbb{Z}^2$			Hypercube $[k]^d$
k	3	$\mathbb{N}\backslash 3\mathbb{N}$	$3\mathbb{N}\backslash\{3\}$	–
min.	$\frac{1}{17}$	$1-\frac{16}{k} \lesssim \bullet \lesssim 1-\frac{8}{k}$		$1-\frac{2d}{k}+\frac{2d(d-1)}{k^2} \pm O(k^{-3})$
max.	$\frac{1}{5}$	$1-\frac{2}{k}$	$1-\frac{2}{k-1} \leq \bullet \leq 1-\frac{2}{k}$	$1-\frac{2}{k}-O(k^{-2})$

Table 1: Main results

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Chapter 1 Introductions

1.1 Monochromatic k in a row

1.1.1 Five in a Row

Five in a Row, also called "Gomoku", is a strategy board game played on a 15×15 grid board (or a 19×19 go board). Two players take turns claiming positions by placing a stone of their color on the board. The winner is the first player to form a consecutive line of length five or more of their color. Once the board is completely filled and no one could claim winning, the game ends in a draw. In different settings, it is possible to require the length of this line to be exactly five, six or more is not allowed. In here, we always allow a longer line. **Monochromatic Five in a Row** is as the same as the original Five in a Row, but all players are using stones of the same color, which means that one can claim a win if their last move forms a line of length five or more on the board. A significant difference to the original game is that this variant never draws. Given $k, t, d \in \mathbb{N}$ and $\{\ell_i\}_{i=1}^{d} \subseteq \mathbb{N}$, an obvious generalization can be obtained: There are t players claiming, in turns, an unclaimed position from a d-dimensional gird board $\prod_{i=1}^{d}[\ell_i]$. The game continues until a player wins or the board is filled. A player wins if there is a k in a row among all claimed positions after their latest claiming. At the end of the game, if no player wins, the board must be filled, then the game is a draw. (It happens when the board is too small to have a k in a row.) We call this game a t-player monochromatic k in a row played on the board $\prod_{i=1}^{d}[\ell_i]$. In this thesis, the winning strategy and the extreme density of stones on the board of 2-player monochromatic variant will be discussed.

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1.1.2 Torus Board

There is nothing special about the board $\prod_{i=1}^{d}[\ell_i]$. In fact, one can find that it is always a draw regardless anything, if we have $\ell_i < k$ for all $i \in [d]$. In general, one can find that the two statements, "the game is a draw", and "the board contains no any k in a row", are equivalent. Note $[k-1]^d$ contains no k in a row, hence a game on a board $\prod_{i=1}^{d}[\ell_i] \subseteq [k-1]^d$ is trivial. A possible modification to make this game non-trivial is replacing each $[\ell_i]$ with $\mathbb{Z}/\ell_i\mathbb{Z}$, then the board becomes $\prod_{i=1}^{d}\mathbb{Z}/\ell_i\mathbb{Z}$, which is called a torus board. From $[\ell_i]$ to $\mathbb{Z}/\ell_i\mathbb{Z}$, it means we can extend a short line reaching the border to a longer line using the modular operation. For example, in $[3] \times [4]$, the set $\{(1,1),(2,2),(3,3)\}$ is a 3 in a row. We correspond it to the line $\{(0,0),(1,1),(2,2)\}$ in $(\mathbb{Z}/3\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})$. Then we can extend it to $\{(0,0),(1,1),(2,2),(0,3),(1,0)\}$, a 5 in a row.

A huge difference between the two versions is that the first player will lose the advantage of "being able to choosing the very first position" since all torus boards (of the same shape) with exactly one position claimed are identical. The focus of this type will be exploring the difference between the toric version and the original version. That is, finding a board on where results of two types differs. Also, characterising whether a winning strategy, for plane board, can be applied to the toric version is concerned.

1.1.3 Infinite Plane Board

If we play the monochromatic k in a row on the infinite plane grid board instead. What can we say? Compare to the normal k in a row on infinite plane board $\mathbb{Z}^2$. When $k = 8$ or 9, we can find a pairing strategy for the second player, hence it is a draw (see p. 159-161 in [1]), while it is unknown when $k = 6$ or 7.

However, the result for the monochromatic ones is worlds apart. Note that you can always claim a position at least $2k$ far from every claimed position, and another player cannot force you to drop this strategy unless creating a win for you. Therefore, it is a trivial draw for the monochromatic game played on infinite board. The statement holds for all

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$k \in \mathbb{N}$ but for $k = 1, 2$, two trivial cases in which we are not interested. Instead, we focus on the behaviors of sequences of k in a row played on $[n]^2$ as $n \to \infty$. Two objectives are concerned: First, to determine which pair (n, k) is a first player win. Second, the bound of densities of finished boards. The former is a classic probe in this field. The latter somehow framed the duration of the game, i.e., how fast/slow can a player end the game. It benefits the study for the biased games.

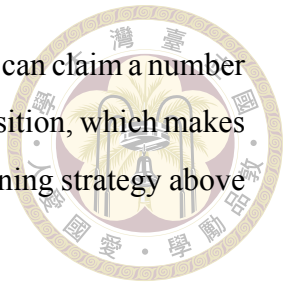
1.1.4 Biased Game

The biased monochromatic k in a row involves two parameters $p, q \in \mathbb{N}$. It is the same as the monochromatic k in a row, except the first player claims p **positions** per move, and the second player claims q **positions** per move, respectively. Denote it the biased $(p : q)$ monochromatic k in a row. The weak biased $(p : q)$ game can be defined in a similar manner, by allowing players to claim lesser positions but forbidding skipping (claiming no position). First, if $p \geq k$, then the first player wins without doubt. Similarly, if $p < k \leq q$, then the second player wins. Now we assume that both of p and q are strictly lesser than k. Suppose $p > q$, the weak $(p : q)$ k-game is a first player win. The following is a possible winning strategy for the first player: (1) Track the minimum number of positions needed to form a k in a row based on current board state, call this number N. (2) When it is the first player's turn, if $N \leq p$, then the first player can claim N positions to win; otherwise, they claims a position that decreases N by one. If the second player wins under this scenario, let S be the board state just before the second player claims their last move. On S, by the definition, the latest move of the first player is a single position x such that $N(S) = N(S \backslash \{x\}) - 1$. As $N(S) \leq q < p$, we know that $N(S \backslash \{x\}) = N(S) + 1 \leq p$. Therefore, the first player could win at the spot. Besides, note that $N(\emptyset) = k$ and the parameter N decreases every other steps, it will eventually become not greater than p in a finite steps. In other words, the game ends in a finite time even if the board is infinite. Similarly, by using the same strategy, when $p < q$, the second player wins the weak $(p : q)$ game. It turns the weak biased game is less interested unless $p = q$.

A remark is that, as long as the board is huge enough, the same results hold for the

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strong versions as well. In a huge board, during the early game, players can claim a number of "dummy" positions that are at least k far from each other claimed position, which makes almost no difference between strong and weak versions. Then the winning strategy above works, still.



1.2 Nikodym & Kakeya Problem

1.2.1 Nikodym Set

The following is a short introduction to Nikodym set and its variant.

Definition 1.2.1. Given a space A with a line structure $L(A) \subseteq 2^A$. A subset $N \subseteq A$ is called a **Nikodym set** if, for every $x \in A \backslash N$, there is a line $L \in L(A)$ such that $x \in L$ and $|L \backslash N| = 1$, i.e., the only missing point on the line L is x.

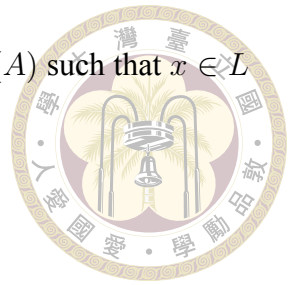
Trivially, A itself is a Nikodym set. For every $x \in A$, the one-point-missing subset $A \backslash \{x\}$ is also a Nikodym set. Note that the smaller a subset is, the harder it is to be a Nikodym set. Therefore, the goal should be to find a Nikodym set that is as small as possible. Li [4] showed that every Nikodym set in $\mathbb{F}_q^n$ has size at least $\binom{q+n-2}{n}$. Especially, the bound can be raised to $\frac{2q^2}{3} + O(q)$ when $n = 2$. Lund, Saraf, and Wolf [6] showed that, for a sufficiently large q, every Nikodym set in $\mathbb{F}_q^3$ has size at least $0.38q^3$.

Later, we will see the connection between the monochromatic k in a row problem and the Nikodym set problem. Roughly speaking, a "finished board state" can be regard as a sort of Nikodym set if we replace "full line" with "line segment of length k" in the definition of Nikodym set. Fortunately, the replacement is needless once every full line, in the definition of Nikodym set, consists of k points. This happens when there is a finite field of size k and the space $A = \mathbb{F}_k^d$. Thus the corresponding games, the monochromatic k in a row played on torus boards, are concerned. In fact, the connection we had is between the final board state and the **2-Nikodym set**, which is a type of generalized Nikodym set defined below.

Definition 1.2.2. Given a space A with a line structure $L(A) \subseteq 2^A$. Let $s \in \mathbb{N}$. A subset

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$N_s \subseteq A$ is s-**Nikodym** if, for every $x \in A \backslash N_s$, there is a line $L \in L(A)$ such that $x \in L$ and $|L \backslash N_s| = s$.



1.2.2 Kakeya Set

The Kakeya set is a related concept to the Nikodym set, and the core problems in the field are the same, asking about how small can a Kakeya set be. There are two types of Kakeya set, that are mainly discussed, based on different underlying spaces. The first one, the classic one, is defined on the real spaces $\mathbb{R}^n$.

Definition 1.2.3 (Kakeya sets in reals)**.** A compact subset $K \subseteq \mathbb{R}^n$ is **Kakeya** if, for every direction $d \in S^{n-1}$, there is a point $x \in \mathbb{R}^n$ such that $x + [0,1]d \subseteq K$, i.e., it contains a **unit line segment** in each direction.

The main conjecture, as known as Kakeya set conjecture, about this type asserts that "every Kakeya set in $\mathbb{R}^n$ has Minkowski and Hausdorff dimension n." Recently, Wang and Zahl [7] proved that the conjecture is true on $\mathbb{R}^3$. The second one is defined on the finite fields. In the finite fields, a Kakeya set is required to contain a full line, instead of a unit line segment, in each direction.

Definition 1.2.4 (Kakeya sets in finite fields)**.** Let $\mathbb{F}_q$ be a finite field of size q. A subset $K \subseteq \mathbb{F}_q^n$ is **Kakeya** if, for every direction $d \in \mathbb{F}_q^n \backslash \{0\}$, there is a point $x \in \mathbb{F}_q^n$ such that $x + (\mathbb{F}_q)d \subseteq K$, i.e., it contains a line in each direction.

Bukh and Chao [2] proved that the size of every Kakeya set $K \subseteq \mathbb{F}_q^n$ satisfies $|K| \geq (2 - \frac{1}{q})^{-(n-1)} q^n$.

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Outline



In the following, we focus on determining the game length and the winner for various settings. For the former, we want results in form of bounds on density of final board states. For the latter, the construction or existence of a winning strategy is needed.

The second chapter consists of the formal definitions (of boards, games, and strategies), handful properties, and problems we are interested in. Notably, the relation between the final board state and the 2-Nikodym set is explained in this chapter. The third chapter is proving main results, including bounds for maximum and minimum densities on two types of boards and a construction of winning strategy for a specific type of boards. Also, some results involving torus board are included. In the final chapter, we give a generalized version of the result mentioned in Section 1.2.1. Besides, we propose several possible extensions from the original problem in different direction and list conjectures we are concerned about.

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Chapter 2 Problem Settings

In general, we can consider the game Monochromatic k in a Row played on a given board B which has an intrinsic k-point structure $L = L(B) \subseteq \binom{B}{k}$ in it, we call such game a (k, B)-game. For instance, $(k, [\ell]^d)$-game is concerned, we take the board $B = [\ell]^d$ along with geometry lines

$$L([\ell]^d) := \big\{\{p, p+r, \cdots, p+(k-1)r\} \subseteq [\ell]^d : r \in \{-1, 0, 1\}^d \backslash \{(0, 0, \cdots, 0)\}\big\}.$$

We call this type the **standard** game. If we replace $[\ell]^d$ by $(\mathbb{Z}/\ell\mathbb{Z})^d$, then we get the **toric** version. The difference of two types is that, to forms a line, "crossing through" the border is allowed in the latter but not in the former.

In fact, there is nothing special for two players since the stones are all of the same color, so we can further define a (k, B, t)-game consisting of t players claiming a position, in turns, on the board B in order to cover some elements in $L(B)$ first. For short, we call a subset l in $L(B)$ a "k**-line**" instead.

2.1 Game Sequence & configuration

In the following chapter, we give the notation describing playing processes and explore the properties of final board states. First, we can regard the playing process as an ordered subset $S = \{s_i\}_{i\in[n]}$ of the board B as follows: In the sequence, for each i, $s_{2(i-1)+1}$ represents the position of the i-th move made by the first player, and $s_{2(i-1)+2}$ represents the position of the i-th move made by the second player. A remark is that, although S is a sequence, we use $\{s_i\}_{i\in[n]}$ instead of $(s_i)_{i\in[n]}$ since we want to emphasize

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that "S is a subset of B" more than the order on the S.



Definition 2.1.1. Given an ordered subset $S = \{s_i\}_{i\in[n]} \subseteq B$, it is called a **game sequence** of the (k, B)-game if

1. S contains a k-line.
2. None of the substrings $S_1^j := \{s_i\}_{i\in[j]}$, for $j \in [n-1]$, contains a k-line.

We call n the **length** of this game sequence, denoted by $\text{len}(S)$. Moreover, we define

$$\text{con}(S) := \{s_i : i \in [n-2]\} = S([n-2]),$$

where we forgo the order on S, and call it the **configuration** produced from S.

Remark 2.1.2. Given a (k, B)-game, suppose $L(B) \neq \emptyset$, then this game never draws, and hence every playing processes can be presented by a game sequence.

Convention 2.1.3. Every board B is assumed to have an non-empty $L(B)$.

Now we give a reasonable assumption on players, that is, a player will claim a position if that position makes a k-line.

Definition 2.1.4. Given a game sequence $S = \{s_i\}_{i\in[n]}$ of (k, B)-game, it is **minimal** if there is no game sequence $S' = \{s'_i\}_{i\in[m]}$ with $m < n$ such that

$$(S')_1^{m-1} = S_1^{m-1},$$

in other words, $s'_i = s_i$ for every $i \in [m-1]$.

It is clear that all game sequences, produced by games in which players follow the assumption, are minimal.

Example 2.1.5. The following are game sequences of $(3, [3]^2)$-game. S is not minimal since there is another shorter game sequence S' such that $(S')_1^4 = S_1^4$.

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1	5	6
4		3
	2	

Figure 2.1: S

1		
4		3
5	2	

Figure 2.2: S'

1		
3		
4	2	

Figure 2.3: S''

In addition, we give another assumption on players, that is, players would avoid creating a **near k-line** if able. (A near k-line means a k-line with a position missed only.)

Definition 2.1.6. Given a minimal game sequence S of the (k, B)-game, it is called **saturated** if there is no minimal game sequence S' such that

$$\text{con}(S) \subsetneq \text{con}(S').$$

A subset $A \subseteq B$ is called a **saturated configuration** if there is a saturated game sequence S such that $\text{con}(S) = A$. The set of all saturated configurations of (k, B)-game is denoted by $\mathfrak{G}(k, B)$.

Under such assumption along with previous assumption, all legal game sequences are saturated since every non-saturated game sequence violates the assumption in the second last step.

Example 2.1.7. Consider the game sequence S'' of $(3, [3]^2)$-game shown above. It is minimal but not saturated since we have

$$\text{con}(S'') = \{(1,3),(2,1)\} \subsetneq \{(1,3),(2,1),(3,2)\} = \text{con}(S'),$$

where S' is a minimal game sequence.

2.2 Property of Saturated Configurations

Lemma 2.2.1. Let $S = \{s_i\}_{i\in[n]}$ be a minimal game sequence. Given a permutation $\sigma \in S_n$, if $n-1$ and n are fixed-points of σ, then $S \circ \sigma$ is a minimal game sequence.

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Proof. It is clear that $S \circ \sigma$ is a game sequence since

$$(S \circ \sigma)([i]) \subseteq S([n-1])$$

contains no k-lines for every $i \in [n-1]$ and $(S \circ \sigma)([n]) = S([n])$ contains a k-line. Suppose it is not minimal, say there is a game sequence $S' = \{s'_i\}_{i\in[m]}$ with $m < n$ such that $(S')_1^{m-1} = (S \circ \sigma)_1^{m-1}$. Let $M - 1 := \max\{\sigma(i) : i \in [m-1]\}$. Note $m - 1 \leq n - 2$ and σ fixes $n-1$ and n, we also have $M - 1 \leq n - 2$, i.e., $M < n$. Define a sequence $S'' = \{s''_i\}_{i\in[M]}$ of length M, where

$$s''_i := \begin{cases} s_i & \text{if } i \in [M-1], \\ s'_m & \text{if } i = M. \end{cases}$$

To make S'' an ordered subset, we just need $s'_m \notin S([M-1])$. If not, then $S([M-1])$ covers $S'([m-1]) \cup \{s'_m\} = S'([m])$ and hence it contains a k-line, which is impossible. Moreover, for each $i \in [M-1]$, we have

$$(S'')_1^i = S_1^i.$$

As $S''([M])$ covers $S'([m])$, we can deduce that S'' is a game sequence. Therefore, the fact $(S'')_1^{M-1} = S_1^{M-1}$ implies that S is not minimal, then the result follows. □

Lemma 2.2.2. For a (k, B)-game, given a subset $X \subseteq B$. If X satisfies

($\star$) For every $x \in B$, the set $X \cup \{x\}$ contains no k-lines.

Then there is a saturated configuration A covering X.

Proof. Let $\mathcal{F}_o$ be the collection of orderings $\phi \in B^{\underline{|B|}}$ with $\phi([|X|]) = X$. For every $\phi \in \mathcal{F}_o$, we define $S(\phi)$ to be the substring $\phi_1^{N(\phi)}$ where

$$N(\phi) := \min\{i : \phi([i]) \text{ contains a } k\text{-line}\}.$$

Such index exists since $\phi([|B|]) = B$ contains k-lines, so $S(\phi)$ is well-defined. Clearly, $S(\phi)$ is a game sequence. Due to the assumptions on X, it is easy to see that $\text{len}(S(\phi)) =$

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$N(\phi) \geq |X| + 2$ for every $\phi \in \mathcal{F}_o$. Consider the family $\mathcal{F}_s := S(\mathcal{F}_o)$, the collection of those game sequences. We claim that there is a minimal game sequence in $\mathcal{F}_s$. Suppose not, pick a S_0 from $\arg\min_{S\in\mathcal{F}_s} \text{len}(S)$. As S_0 is not a minimal game sequence, there is another game sequence $S' = \{s'_i\}_{i\in[m]}$ of length $m < \text{len}(S_0)$ such that

$$(S')_1^{m-1} = (S_0)_1^{m-1}.$$

Consider two cases:

(i) If $m - 1 \leq |X|$, then we have

$$\begin{aligned} S'([m]) &= S'([m-1]) \cup \{s'_m\} \\ &= S_0([m-1]) \cup \{s'_m\} \subseteq S_0([|X|]) \cup \{s'_m\} = X \cup \{s'_m\}. \end{aligned}$$

Being a superset of $S'([m])$, as $S'([m])$ contains a k-line, $X \cup \{s'_m\}$ also contains a k-line, which is impossible due to the assumption $(\star)$ on X.

(ii) If $m - 1 > |X|$, we have

$$S'([m]) \supseteq S'([m-1]) = S_0([m-1]) \supseteq S_0([|X|]) = X.$$

Then it is possible to pick an ordering ϕ' of B satisfying

$$\phi'([|X|]) = X \quad \text{and} \quad \phi'([m]) = S'([m]).$$

The former implies $S(\phi') \in \mathcal{F}_s$, and the latter implies $\text{len}(S(\phi')) = N(\phi') \leq m$ since $S'([m])$ contains a k-line. Hence we have $\text{len}(S(\phi')) \leq m < \text{len}(S_0)$, a contradiction.

Let $\mathcal{F}_{ms}$ be the collection consisting of all minimal game sequences in $\mathcal{F}_s$. Pick an arbitrary game sequence S_1 form $\arg\max_{S\in\mathcal{F}_{ms}} \text{len}(S)$ and fix it. We claim that S_1 is saturated. Suppose not, there is a minimal game sequence S'' such that

$$\text{con}(S_1) \subsetneq \text{con}(S'').$$

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Recall that $\mathrm{len}(S_1) \geq |X| + 2$, we have

$$\mathrm{con}(S_1) = S_1([N(\phi_1) - 2]) \supseteq S_1([|X|]) = X.$$

In particular, we have $X \subseteq \mathrm{con}(S'')$. Let $\ell := \mathrm{len}(S'')$. Pick a permutation $\sigma \in S_{[\ell]}$ such that $\sigma(\ell - 1) = \ell - 1$, $\sigma(\ell) = \ell$, and

$$(S'' \circ \sigma)([|X|]) = X.$$

By Lemma 2.2.1, $S'' \circ \sigma$ is also a minimal game sequence, hence $S'' \circ \sigma \in \mathcal{F}_{ms}$. However, thanks to the fact $|\mathrm{con}(S'')| > |\mathrm{con}(S_1)|$, we can see that

$$\mathrm{len}(S'' \circ \sigma) = \mathrm{len}(S'') = |\mathrm{con}(S'')| + 2 > |\mathrm{con}(S_1)| + 2 = \mathrm{len}(S_1),$$

which contradicts to the fact that S_1 belongs in $\arg\max_{S \in \mathcal{F}_{ms}} \mathrm{len}(S)$. Eventually, we set $A := \mathrm{con}(S_1)$. Note that, by the definition of $\mathcal{F}_o$, $\mathrm{con}(S_1) \supseteq S([|X|]) = X$, then the result follows. □

Property 2.2.3. For every (k, B)-game, we have

(i) $\mathfrak{G}(k, B)$ is an anti-chain.

(ii) Given a subset $A \subseteq B$ and an arbitrary position $x \in B$, if A is a configuration of some minimal game sequences, then the set $A \cup \{x\}$ contains no k-lines.

(iii) Given a saturated configuration A and an arbitrary position $x \in B \backslash A$, there is a position $y \in B \backslash (A \cup \{x\})$ such that $A \cup \{x, y\}$ contains a k-line.

Proof.

(i) If not, there are two saturated configurations $A_1, A_2 \in \mathfrak{G}(k, B)$ with $A_1 \subseteq A_2$. Let S_i be a saturated game sequence such that $\mathrm{con}(S_i) = A_i$ for $i = 1, 2$. As S_1 is a saturated game sequence, S_2 can not be minimal, which leads to a contradiction. □

(ii) Suppose not, there is a subset $A \subseteq B$ satisfying that condition and a position $x \in B$ such that $A \cup \{x\}$ contains a k-line. We want to find a game sequence S' inside $A \cup \{x\}$. Pick an fix an arbitrary order σ on A. Consider two cases:

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(a) If $x \in A$. Define an ordered subset $S'' := \{\sigma(i)\}_{i\in[|A|]}$. Let i_0 be the smallest index such that $S''([i_0])$ contains a k-line. Such index exists since $S''([|A|]) = A = A \cup \{x\}$ contains a k-line. Then we set $S' := (S'')_1^{i_0}$.

(b) If $x \notin A$. The ordered subset $S'' = \{s''_i\}_{i\in[|A|+1]}$ is defined, where

$$s''_i := \begin{cases} \sigma(i) & \text{if } \ i \in [|A|], \\ x & \text{if } \ i = |A| + 1. \end{cases}$$

Similar to previous, let i_0 be the smallest index such that $S''([i_0])$ contains a k-line. The existence of i_0 is ensured by the fact that $S''([|A| + 1]) = A \cup \{x\}$ contains a k-line. Then we set $S' := (S'')_1^{i_0}$. An observation is that, in this case, $x \in S'$ if and only if $i_0 = |A| + 1$. In particular, $S'([i_0 - 1]) \subseteq A$.

Let $S = \{s_i\}_{i\in[|A|+2]}$ be a minimal game sequence with $\mathrm{con}(S) = A$. Note that, in either cases, we have

$$S'([i_0 - 1]) \subseteq A = S([|A|]) = S([|A| + 2])\backslash\{s_{|A|+1}, s_{|A|+2}\}.$$

By Lemma 2.2.1, we may assume that $S_1^{i_0-1} = (S')_1^{i_0-1}$. As we can see that $\mathrm{len}(S') = i_0 \leq |A| + 1 < |A| + 2 = \mathrm{len}(S)$, S is not a minimal game sequence, which leads to a contradiction. □

(iii) Given a saturated configuration $A \in \mathfrak{G}(k, B)$. Let $S = \{s_i\}_{i\in[n]}$ be a game sequence such that $\mathrm{con}(S) = A$. Suppose the claim is not true, there is a position $x \in B\backslash A$ such that, for every $y \in B\backslash(A \cup \{x\})$, $A \cup \{x, y\}$ contains no k-lines. In particular, the set $A \cup \{x\}$ must contain no k-lines. Apply Lemma 2.2.2 on $A \cup \{x\}$, we get a saturated configuration A' with $A' \supseteq A \cup \{x\} \supsetneq A$, which contradicts to the fact that $\mathfrak{G}(k, B)$ is an anti-chain. □

So far, we can tell that the concept of saturated configurations is sophisticated and hard to manipulate. The main goal of this part is to develop an equivalence that is easy to understand and/or verify.

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Definition 2.2.4. Given a (k, B)-game. For a subset $A \subseteq B$, if A satisfies

$$[\forall x \in B, A \cup \{x\} \text{ contains no } k\text{-lines.}]$$

Then we call it a **minimal configuration** of (k, B)-game. The collection of all minimal configurations is denoted by $\mathfrak{F}(k, B)$.

The definition of minimal configuration does not rely on the game sequences. It is easy to check whether a subset is a minimal configuration or not once we fully know the k-line structure on the board. The relation between saturated configurations and minimal configurations will be unveiled in the following theorem.

Theorem 2.2.5. Given a (k, B)-game. For every subset $A \subseteq B$, the following are equivalent:

(i) A is a saturated configuration.

(ii) A is a minimal configuration and is maximal under set inclusion in $\mathfrak{F}(k, B)$.

(iii) A is a minimal configuration, and, for every $x \in B \backslash A$, there is a $y \in B \backslash (A \cup \{x\})$ such that $A \cup \{x, y\}$ contains a k-line.

Proof.

$(i) \implies (iii)$ Note that A is the configuration of some minimal game sequences, according to the second property from 2.2.3 and the definition of minimal configuration, the subset A is a minimal configuration. Further, by the third property from 2.2.3, A satisfies the rest condition in (iii). □

$(iii) \implies (ii)$ It is suffice to show that A is maximal in $\mathfrak{F}(k, B)$. Suppose not, there is a minimal configuration A' such that $A' \supsetneq A$. Pick and fix a $x \in A' \backslash A \neq \emptyset$. By (iii), there is a position $y \in B \backslash (A \cup \{x\})$ such that $A \cup \{x, y\}$ contains a k-line. As A' is a minimal configuration, the set $A' \cup \{y\}$ contains no k-lines. Therefore, the containment

$$A \cup \{x, y\} \subseteq A' \cup \{y\}$$

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leads to a contradiction. □

$(ii) \implies (i)$ Suppose A is a minimal configuration but not a saturated configuration. Then there is a minimal game sequence $S = \{s_i\}_{i\in[n]}$ satisfying $A \subsetneq \text{con}(S)$. Thanks to the second property from 2.2.3, we know $\text{con}(S)$ is a minimal configuration, i.e., $\text{con}(S) \in \mathfrak{F}(k, B)$. As a result, A is not maximal in $\mathfrak{F}(k, B)$. □

Let A be a saturated configuration, for every $x \in B\backslash A$, let $R(x)$ denote the collection of all possible choices for y satisfying (iii) in the previous theorem. Thanks to the theorem and Lemma 2.2.1, we have the characterization of all possible saturated game sequences

$$\{S : S \text{ is a saturated game sequence with con}(S)\text{=A}\}$$
$$= \{S = \{\sigma(1), \sigma(2), \cdots, \sigma(|A|), x, y\} : \sigma \in A^{\underline{|A|}}, x \in B\backslash A, y \in R(x)\}.$$

With this identification, the study of saturated configurations becomes the foundation of this problem. In order to make the definition more sensible, we give a generalization of the conditions for saturated configurations.

Definition 2.2.6. Given a subset $A \subseteq B$. For every k-line $L \in L(B)$, we say L is a **saturated line** of A (or L is saturated in A) if

$$|L\backslash A| = 2,$$

and it is called a **over-saturated** line of A if

$$|L\backslash A| < 2.$$

Definition 2.2.7. Given a subset $A \subseteq B$. For every position $x \in B$, we say x is **suppressed** by a k-line $L \in L(B)$ if

$$x \in L\backslash A,$$

and L is a saturated line of A.

Observation 2.2.8. Given a (k, B)-game and a subset $A \subseteq B$. A is a saturated configuration if and only if there is no over-saturated line of A and every position in $B\backslash A$ is

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suppressed by some saturated lines of A.



Note that the second condition "every position outside A is suppressed" means

$$\forall x \in B \backslash A \; \exists L \in L(B) \text{ s.t. } x \in L \text{ and } |L \backslash A| = 2,$$

which is the same as the definition of 2-Nikodym set if we regard $L(B)$ as the line structure when defining Nikodym set on B. Hence any result about 2-Nikodym set also holds for saturated configuration. Besides, there is an additional global constraint, "being 'over-saturated line'-free", applying on saturated configurations. That is

$$\forall L \in L(B), \; |L \backslash A| \geq 2.$$

Then we may define the generalization:

Definition 2.2.9. Given a set B along with a non-empty line structure $L(B) \subseteq 2^B$. Let $s \in \mathbb{N}$. A subset $A \subseteq B$ is said to be s**-saturated** if A is a s-Nikodym set and satisfies

$$\forall L \in L(B), \; |L \backslash A| \geq s.$$

2.3 Density of Configurations

There are plenty objects we are interested in, one is how "long" the game will last. However, the duration of games varies and mainly depends on the board size. For instance, a $(3, [3]^2)$-game has at most $|[3]^2| = 9$ steps. In contrast, a $(3, [125]^2)$-game needs at least 625 steps since, when the game ends, every 5 by 5 square on the board can not be empty. As the board $[125]^2$ can be partitioned into 625 such squares, the claim follows. Therefore, we decide to measure the ratio of the number of steps to the board size, instead of the number of steps. As the ratio falls in the interval $[0, 1]$, the comparison between games becomes easier.

Definition 2.3.1. Given a (k, B)-game with a finite board. For every subset $A \subseteq B$, its

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density is defined to be

$$\frac{|A|}{|B|},$$

and is denoted by $d(A)$. The **density spectrum** of this game is the subset

$$\mathcal{D}(k,B) := \{d(S) : S \in \mathfrak{G}(k,B)\} \subseteq [0,1]_{\mathbb{R}}.$$

Further, the **minimum density** and **maximum density** of this game are

$$d(k,B) := \min \mathcal{D}(k,B) \text{ and } D(k,B) := \max \mathcal{D}(k,B),$$

respectively.

Remark. In fact, the actual density spectrum of (k,B)-game should be

$$\mathcal{D}'(k,B) := \{d(S) : S \subseteq B, S \text{ is a saturated game sequence of } (k,B)\text{-game}\}.$$

Fortunately, there is correspondence between two families: saturated game sequences and saturated configurations. Given a saturated game sequence S, we know $\text{con}(S) \in \mathfrak{G}(k,B)$. In the other hand, we know, for every saturated configuration $A \in \mathfrak{G}(k,B)$, there is a saturated game sequence S such that $\text{con}(S) = A$. Note that, for every game sequence S, we have $|S| = |\text{con}(S)| + 2$. Hence we have

$$\mathcal{D}'(k,B) = \mathcal{D}(k,B) + \frac{2}{|B|},$$

i.e., the spectrum we use is shifted by a constant $\frac{2}{|B|}$. As $|B|$ grows, the error between two spectrum can be ignored. As long as we have fully characterized saturated configurations, we focus on $\mathcal{D}$ instead of $\mathcal{D}'$.

There are two types of games we mainly concern: First, the games played on the whole plane $\mathbb{Z}^2$. In other words, the $(k,\mathbb{Z}^2)$-game for every $k \in \mathbb{N}$. We call this type the **plane games**. Second, the monochromatic k in a row played on the hypercube with side length k and dimension d. That is, the $(k,[k]^d)$-game for every $k,d \in \mathbb{N}$. This type is called the **hypercube games**.

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Notably, those games are trivial when $k = 1$ or 2. For convenience, we assume that $k \geq 3$. One may observe that our definition of densities does not work on the plane game family since their board, $\mathbb{Z}^2$, is not finite.

Definition 2.3.2. For the plane games $\{(k, \mathbb{Z}^2)\}_{k\in\mathbb{N}}$, the **minimum density** and **maximum density** are defined to be

$$d(k, \mathbb{Z}^2) := \lim_{\ell\to\infty} d(k, [\ell]^2) \text{ and } D(k, \mathbb{Z}^2) := \lim_{\ell\to\infty} D(k, [\ell]^2),$$

respectively.[1]

2.4 Strategies

Being a positional game, one of the most common quests is determining the winning player. A niche note is that the monochromatic k in a row is not a strong game[2], which means that having more positions claimed is not always better. Thus we can not directly apply the results from the field of strong positional games.

Definition 2.4.1. Given a (k, B)-game. Let $\mathcal{F} = \mathcal{F}(B) \subseteq 2^B$ be the family of all subsets containing no k-lines, every subset in $\mathcal{F}$ is called a **possible board state**. A map $\phi : \mathcal{F} \to B$ is called a **strategy** of (k, B)-game if $\phi(A) \notin A$ for all $A \in \mathcal{F}$.

The image $\phi(S)$ is the position that a player using the strategy ϕ would claim when they see the board state $S \in \mathcal{F}$. That is the reason why we only need to define the strategy on $\mathcal{F}$ since the game ends once the board state leaves the family $\mathcal{F}$. In fact, we can chop down the domain into two parts, the collection of odd subsets in $\mathcal{F}$ and the collection of even subsets in $\mathcal{F}$. We denote them $\mathcal{F}_o$ and $\mathcal{F}_e$, respectively. One can find that a strategy for the first player only cares the images of $\mathcal{F}_o$, only. Similarly, a strategy for the second player cares the images of $\mathcal{F}_e$, only. For convenience, we extend the range B to $B \cup \{\star\}$. We set $\phi(S) = \star$ if S is not a possible board state that player would see while playing against any strategies. For instance, if ϕ is a strategy for the first player, then we can set

[1] The limits might depend on how $\mathbb{Z}^2$ is approached. Here we use squares $[\ell]^2$ to approach.
[2] Strong games is a type of position games, one can find the definition from Chapter 1 of [3].

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$\phi(S) = \star$ for every $S \in \mathcal{F}$ with $\phi(\emptyset) \notin S$ since $\phi(\emptyset)$ is the first position that the first player claimed.

Definition 2.4.2. Given two strategies ϕ_1 and ϕ_2 of (k, B)-game. First, we set $s_1 := \phi_1(\emptyset)$. Iteratively, for each $i \in \mathbb{N}$, we define

$$s_i := \begin{cases} \phi_1(\{s_j\}_{j\in[i-1]}) \text{ if } i \text{ is odd,} \\ \phi_2(\{s_j\}_{j\in[i-1]}) \text{ if } i \text{ is even} \end{cases}$$

if $\{s_j\}_{j\in[i-1]} \in \mathcal{F}$. Let n be the index that $\{s_i\}_{i\in[n-1]} \in \mathcal{F}$ but $\{s_i\}_{i\in[n]} \notin \mathcal{F}$. Then the game sequence $S(\phi_1, \phi_2) := \{s_i\}_{i\in[n]}$ is called the **game process** of ϕ_1 and ϕ_2.

Definition 2.4.3. Given a strategy ϕ, we say:

1. It is **first-player-winning**[3] **(FP-winning)** if, for every strategy ϕ', the game process $S(\phi, \phi')$ is of odd length.

2. It is **second-player-winning (SP-winning)** if, for every strategy ϕ', the game process $S(\phi', \phi)$ is of even length.

In both cases, it is called a **winning strategy**.

Note that, for any game, the FP-winning strategy and the SP-winning strategy can not exist at the same time. Otherwise, say there is a (k, B)-game having both of FP-winning and SP-winning strategy. Let ϕ and ϕ' be a FP-winning and SP-winning strategy of (k, B)-game, respectively. Then, by the definition, the length of $S(\phi, \phi')$ should be odd and even, which is impossible. In particular, a winning strategy is either FP-winning or SP-winning.

Definition 2.4.4. Given a (k, B)-game. If there is a FP-winning strategy of (k, B)-game, this game is said to be of **type FP**; otherwise, it is said to be of **type SP**.

Fact 2.4.5. Every (k, B)-game is of either type **FP** or type **SP**.[3]

[3]The "FP" and "SP" notations follow the section 1 in Chapter 2 of [3]

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Chapter 3 Main Results

3.1 Plane Game

3.1.1 Bounds for $d(k, \mathbb{Z}^2)$: Plane Tiling

3.1.1.1 Lower Bound: Line-Space Correspondence

Theorem 3.1.1. The inequality

$$d(k, \mathbb{Z}^2) \geq \frac{k-2}{16+(k-2)}$$

holds for every $k \geq 3$.

Proof. Let S be a saturated configuration of $(k, [\ell]^2)$-game. Let $N := [\ell]^2 \backslash S$. Consider a bipartite graph $G = (N \sqcup S, E)$ defined by

$$E := \{\{x, y\} : x \in N,\ y \in S,\ \text{there is a saturated line containing } x \text{ and } y\}.$$

Given a position $x \in N$, by the property of the saturated configuration, we know there is a saturated line $L \in L([\ell]^2)$ containing x, hence $\delta(N) \geq k-2$. Conversely, given a position $y \in S$, for each direction $\vec{d} \in \{-1, 0, 1\}^2 \backslash \{(0,0)\}$, we claim that there are up to two neighbours of y in the direction $\vec{d}$ from y. To show this, we order the positions in $(y + \mathbb{N}\vec{d}) \cap N$ following the direction $\vec{d}$, say $(y + \mathbb{N}\vec{d}) \cap N = \{n_i\}_{i \in [m]}$ for some $m \in \mathbb{N}$. Note that, for each $i \geq 3$, any k-line containing both y and n_i contains at least $i \geq 3$ positions from N, which is thus non-saturated. As a result, there are up to two neighbours, n_1 and n_2, in the direction $\vec{d}$ from y.

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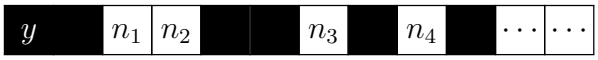


Figure 3.1: The ordering on $(y + \mathbb{N}\vec{d}) \cap N$, the direction $\vec{d}$ goes right.

Therefore, as there are eight directions in total, we have $\Delta(S) \leq 16$. It leads to the following inequality

$$(k-2)|N| = \delta(N)|N| \leq |E(G)| \leq \Delta(S)|S| \leq 16|S|.$$

Adding $(k-2)|S|$ on both sides, we get $(k-2)\ell^2 \leq (16 + (k-2))|S|$ and hence

$$\frac{|S|}{\ell^2} \geq \frac{k-2}{16 + (k-2)}.$$

Let S_ℓ be a saturated configuration of $(k, [\ell]^2)$-game such that $\frac{|S_\ell|}{\ell^2} = d(k, [\ell]^2)$, then

$$d(k, \mathbb{Z}^2) = \lim_{\ell \to \infty} d(k, [\ell]^2) = \lim_{\ell \to \infty} \frac{|S_\ell|}{\ell^2} \geq \frac{k-2}{16 + (k-2)}.$$

□

3.1.1.2 Upper Bound: Pattern Tiling

Theorem 3.1.2. For $k \in \mathbb{N}_{\geq 4}$, we have

$$d(k, \mathbb{Z}^2) \leq \frac{(k-2)^2 + 8}{(k+2)^2}.$$

Proof. Consider a square of side length $k-2$ on the plane. If all positions in it are claimed, then surrounding $8k - 8$ positions (colored in gray) are suppressed. By claiming the 8 positions nearing corners, we call the $k + 2$ by $k + 2$ square a **filled-cog**, and call the central $k - 2$ by $k - 2$ square the **core** of this cog.

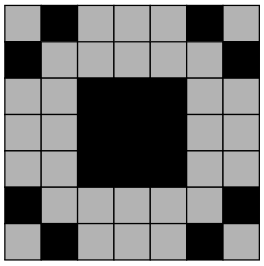


Figure 3.2: Filled-cog for $k = 5$. ■: suppressed by core.

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By tiling up such pattern, we get the desired upper bound.[1] □

Although adding 8 positions into the pattern has asymptomatically small effect on the density, this trick does not work well for small k. However, in such case, we can take another modification on the pattern to get a better bound.

Corollary 3.1.3. For $k \in \mathbb{N}_{\geq 3}$, we have

$$d(k, \mathbb{Z}^2) \leq \frac{(k-2)^2}{k^2+2k+2}.$$

Proof. Like the proof of Theorem 3.1.2, we use the filled-cog but leaving the 8 positions nearing the corners unclaimed, and call such pattern a **cog**.

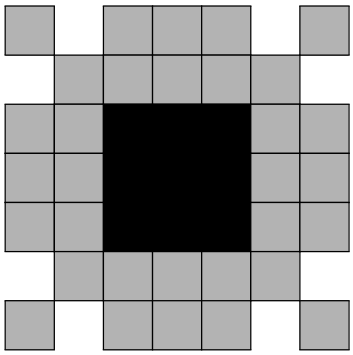


Figure 3.3: Cog for $k = 5$

Suppose we have a collection of cogs such that their union covers the plane. Let S be the union of cores. To make S a configuration, we also need that, for every core, it does not intersect with other cogs. To minimize the density of S, we need to minimize the number of cogs used to cover $[\ell]^2$ for almost all $\ell \in \mathbb{N}$. Intuitively, this can be done by minimizing the size of overlapping area among cogs. In particular, we take the following covering.

[1] Note that the corner positions are near 3-lines, hence this construction does not work for $k = 3$.

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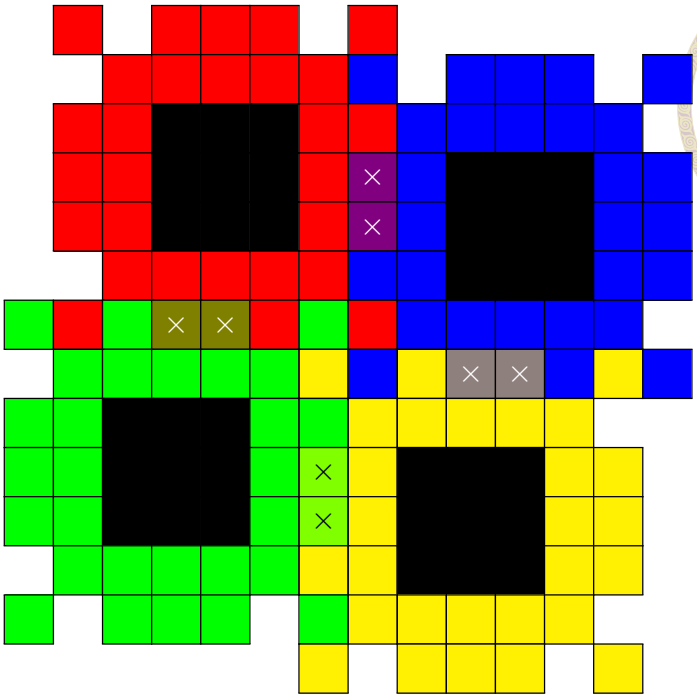


Figure 3.4: Shifted cog covering ($\times$: overlapping)

Equivalently, we can remove the positions in the top-most row or the left-most column intersecting with other pieces, call it the **modified-cog**. We can use this new structure to tile the plane in the following way.

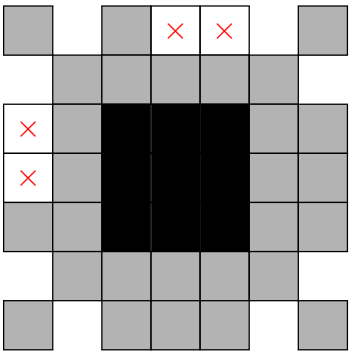


Figure 3.5: Modified-Cog for $k = 5$

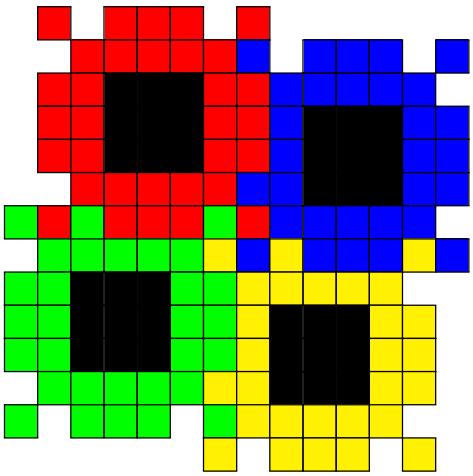


Figure 3.6: Shifted modified-cog tiling

In average, each core suppresses $6k - 2 + O_\ell(\frac{k^2}{\ell})$ unclaimed positions. And there are at most $4k\ell$ uncovered positions under this tiling in $[\ell]^2$, which raises the number of

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claimed positions in each cog by $O_\ell(\frac{k^3}{\ell})$. It turns we have the upper bound

$$d(k,\mathbb{Z}^2) \leq \lim_{\ell\to\infty} \frac{|\text{core}| + O(\ell^{-1})}{|\text{core}| + 6k - 2 + O(\ell^{-1})} = \frac{(k-2)^2)}{(k-2)^2 + 6k - 2} = \frac{(k-2)^2}{k^2+2k+2}.$$

□

Remark. The adjusted bound from Corollary 3.1.3 outperforms the original one from Theorem 3.1.2 when $k \leq 8$, and works for the case $k = 3$.

3.1.1.3 Tight Bound

Due to Theorem 3.1.1 and Corollary 3.1.3, we have

$$\frac{1}{17} = \frac{k-2}{16+(k-2)} \leq d(3,\mathbb{Z}^2) \leq \frac{(k-2)^2}{k^2+2k+2} = \frac{1}{17}.$$

For other k, due to Theorem 3.1.1 and Theorem 3.1.2, we have

$$1 - \frac{16}{k} \lesssim \frac{k-2}{16+(k-2)} \leq d(k,\mathbb{Z}^2) \leq \frac{(k-2)^2+8}{(k+2)^2} \lesssim 1 - \frac{8}{k}.$$

3.1.2 Bounds for $D(k,\mathbb{Z}^2)$: Framing

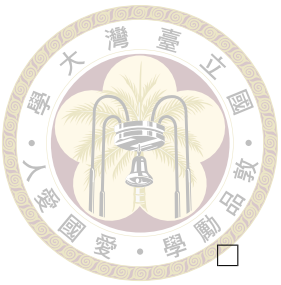
3.1.2.1 Lower Bound: Locus Removal

Theorem 3.1.4. For $k \in \mathbb{N}_{\geq 2}$, we have

$$D(k+1,\mathbb{Z}^2) \geq D(k,\mathbb{Z}^2).$$

Proof of Theorem 3.1.4. We claim that the inequality $D(k+1,[\ell]^2) \geq D(k,[\ell]^2)$ holds for each $k \in \mathbb{N}_{\geq 2}$ and each $\ell \in \mathbb{N}_{\geq k+1}$. Fix a pair of k,ℓ. Let S be a saturated configuration of $(k,[\ell]^2)$-game that is of maximum density. Since S contains no k-line, it contains no $(k+1)$-line. For the same reason, for each $x \in [\ell]^2 \backslash S$, the set $S \cup \{x\}$ contains no $(k+1)$-line. By Theorem 2.2.5, there is a saturated configuration S' of $(k+1,[\ell]^2)$-game such

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that $S' \supseteq S$. Then we have

$$D(k+1, [\ell]^2) \geq \frac{|S'|}{\ell^2} \geq \frac{|S|}{\ell^2} = D(k, [\ell]^2),$$

which implies $D(k+1, \mathbb{Z}^2) \geq D(k, \mathbb{Z}^2)$ as ℓ tends to infinity. □

Theorem 3.1.5. For $k \in \mathbb{N}_{\geq 2}$ with $3 \nmid k$, we have

$$D(k, \mathbb{Z}^2) \geq 1 - \frac{2}{k}.$$

Proof of Theorem 3.1.5. For every $t \in \mathbb{N}_{\geq 2}$ and $r \in \mathbb{Z}/t\mathbb{Z}$, we define

$$Z_t(r) := \{\, \vec{x} \in \mathbb{Z}^2 \ : \ \langle (1,2), \vec{x} \rangle = r \pmod{t} \}.$$

Given $k \in \mathbb{N}_{\geq 2}$ with $3 \nmid k$. In the board $[\ell]^2$, consider the subset

$$S_{\ell,k} := [\ell]^2 \backslash (Z_k(0) \sqcup Z_k(1)).$$

We claim that $|L \cap (Z_k(0) \sqcup Z_k(1))| = 2$ for every k-line L on the board $[\ell]^2$. Given a line L, we have $L = \{\vec{v} + x\vec{d} : x \in [0, k-1]\}$ for some position $\vec{v} = (v_1, v_2) \in [\ell]^2$ and some direction $\vec{d} = (d_1, d_2) \in \{-1, 0, 1\}^2 \backslash \{0\}$. Note that $\vec{v} + x\vec{d} \in Z_k(r)$ if and only if

$$(d_1 + 2d_2)x = r - (v_1 + 2v_2) =: c \pmod{k}.$$

If k is coprime with $d_1 + 2d_2$, there is a unique solution to the above linear equation for every $c \in \mathbb{Z}$. In this case, we have

$$|L \cap Z_k(0)| = |L \cap Z_k(1)| = 1,$$

and hence

$$|L \cap (Z_k(0) \sqcup Z_k(1))| = 2.$$

Note that all possible values of $d_1 + 2d_2$ are $\pm 1, \pm 2, \pm 3$. By the assumption on k, k is already coprime with ± 1 and ± 3. The remaining case is $k \in 2\mathbb{N}$ and $d_1 + 2d_2 = \pm 2$. In

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this case, the number of solutions to the equation is

$$\begin{cases} 2 & \text{if } c \text{ is even,} \\ 0 & \text{if } c \text{ is odd.} \end{cases}$$

It implies that, depending on the parity of $v_1 + 2v_2$, either

$$|L \cap Z_k(0)| = 2, \text{ and } |L \cap Z_k(1)| = 0$$

or

$$|L \cap Z_k(0)| = 0, \text{ and } |L \cap Z_k(1)| = 2$$

holds. In both cases, we have $|L \cap (Z_k(0) \sqcup Z_k(1))| = 2$.

With the knowledge $|L \cap S_{\ell,k}| = k-2$ for every line L on the board $[\ell]^2$, $S_{\ell,k}$ becomes a saturated configuration. It remains to determine the size of $S_{\ell,k}$. One can partition the board $[\ell]^2$ into a collection of lines, $\mathcal{S}$, along with a square X with side length lesser than k. By the same knowledge, we know that

$$\begin{aligned} |S_{\ell,k}| &= \sum_{L \in \mathcal{S}} |L \cap S_{\ell,k}| + |X \cap S_{\ell,k}| \\ &\geq \sum_{L \in \mathcal{S}} k - 2 = \sum_{L \in \mathcal{S}} |L| \frac{k-2}{k} = \left| \bigcup_{L \in \mathcal{S}} L \right| \frac{k-2}{k} \\ &= |[\ell]^2 \backslash X| \frac{k-2}{k} \geq (\ell^2 - (k-1)^2) \frac{k-2}{k}. \end{aligned}$$

It turns that the inequality

$$D(k, \mathbb{Z}^2) \geq \lim_{\ell \to \infty} \frac{|S_{\ell,k}|}{\ell^2} \geq \lim_{\ell \to \infty} \frac{(\ell^2 - (k-1)^2)(k-2)}{\ell^2 k} = 1 - \frac{2}{k}$$

holds for every $k \in \mathbb{N}_{\geq 2}$ with $3 \nmid k$. □

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3.1.2.2 Upper Bound



Theorem 3.1.6. The inequality

$$D(k, \mathbb{Z}^2) \leq 1 - \frac{2}{k}$$

holds for every $k \geq 2$.

Lemma 3.1.7 (Blurring lemma)**.** Given a non-empty finite subset $T \subseteq \mathbb{Z}^2$. There is a constant $C = C(T) > 0$ such that, for every $\ell \in \mathbb{N}$ and every $S \subseteq [\ell]^2$, the following two inequalities

$$\sum_{x \in [\ell]^2} |(T + x) \cap S| \leq |T||S|,$$

and

$$|T||S| - \sum_{x \in [\ell]^2} |(T + x) \cap S| \leq C\ell$$

hold.

Proof. Note that

$$\begin{aligned}\sum_{x \in [\ell]^2} |(T + x) \cap S| &= \sum_{x \in [\ell]^2} \sum_{y \in T} \chi_{x+y \in S} = \sum_{y \in T} \sum_{x \in [\ell]^2} \chi_{x+y \in S} \\ &= \sum_{y \in T} |([\ell]^2 + y) \cap S| \leq \sum_{y \in T} |S| = |T||S|.\end{aligned}$$

Further, we have

$$\begin{aligned}|T||S| - \sum_{x \in [\ell]^2} |(T + x) \cap S| &= \sum_{y \in T} |[\ell]^2 \cap S| - |([\ell]^2 + y) \cap S| \\ &\leq \sum_{y \in T} |[\ell]^2 \backslash ([\ell]^2 + y)| \\ &\leq \sum_{y \in T} \|y\|_1 \ell \leq |T| \max_{y \in T} \|y\|_1 \ell.\end{aligned}$$

Take $C = |T| \max_{y \in T} \|y\|_1$, then the result follows. □

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Definition 3.1.8. Fix a finite subset $T \subseteq \mathbb{Z}^2$ with $0 \in T$, define a map

$$\begin{aligned} n_T : \mathbb{Z}^2 \times 2^{\mathbb{Z}^2} &\to \quad \mathbb{N} \cup \{0\} \\ (x, S) &\mapsto |(T + x) \cap S| \end{aligned}$$

where $n_T(x, S)$ is called the **local number** of S around x based on the **frame** T. The **local density** of S around x (based on the frame T) is defined to be

$$d_T(x, S) := \frac{n_T(x, S)}{|T|}.$$

Corollary 3.1.9. Fix a finite subset $T \subseteq \mathbb{Z}^2$ with $0 \in T$. Pick a sequence of subsets $\{S_\ell\}_{\ell \in \mathbb{N}}$ such that $S_\ell \subseteq [\ell]^2$. Consider the average local density of S_ℓ on $[\ell]^2$

$$\overline{d}_\ell := \frac{1}{\ell^2} \sum_{x \in [\ell]^2} d_T(x, S_\ell).$$

Suppose the sequence $\frac{|S_\ell|}{\ell^2}$ converges to d, for some $d \in \mathbb{R}$, as ℓ goes to infinity, then

$$\lim_{\ell \to \infty} \overline{d}_\ell = d,$$

i.e., the average local density of S_ℓ on $[\ell]^2$ also converges to the same limit d.

Proof. Note that, by Blurring lemma, we have

$$\left| |T||S_\ell| - |T|\ell^2 \overline{d}_\ell \right| = \left| |T||S_\ell| - \sum_{x \in [\ell]^2} |(T + x) \cap S_\ell| \right| = O(\ell).$$

It turns

$$\left| \frac{|S_\ell|}{\ell^2} - \overline{d}_\ell \right| = \frac{O(\ell)}{|T|\ell^2} = O(\ell^{-1}) = o(1).$$

□

Fix a $T \subseteq \mathbb{Z}^2$ with $0 \in T$. Pick a sequence of maximum saturated configurations

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$S_\ell \in \mathfrak{G}(k, [\ell]^2)$. If we have the average local numbers

$$\overline{n}_\ell := \frac{1}{\ell^2} \sum_{x \in [\ell]^2} n_T(x, S_\ell)$$

are smaller than $\alpha + o(1)$ for some $\alpha > 0$, then, by the corollary, $D(k, \mathbb{Z}^2) \leq \frac{\alpha}{|T|}$.

It shows that, in order to bound the maximum density of the plane game, we can try to give an upper bound of average local number, based on a given frame, instead. In contrast to build a maximum configuration, with the help of local restrictions on saturated configurations, it is way more easier to bound the local numbers.

Proof of Theorem 3.1.6. Let the frame T be $\{(0, i) : i \in \mathbb{Z}, 0 \leq i < k\}$, a vertical k-line containing the origin. Given a maximum saturated configuration $S_\ell \in \mathfrak{G}(k, [\ell]^2)$. For each $x \in [\ell] \times [\ell - k + 1]$, the subset $T + x$ is fully contained in the board $[\ell]^2$, i.e., $T + x \in L([\ell]^2)$. Hence, by the definition of saturated configuration, $|S_\ell \cap (T + x)| \leq k - 2$. As a result, the average local number is bounded by

$$\begin{aligned}\overline{n}_\ell = \frac{1}{\ell^2} \sum_{x \in [\ell]^2} n_T(x, S_\ell) &\leq \frac{1}{\ell^2} \left(\sum_{x \in [\ell] \times [\ell - k + 1]} k - 2 + \sum_{x \in [\ell] \times [\ell - k + 2, \ell]} k \right) \\ &= k - 2 + \frac{2|[\ell] \times [\ell - k + 2, \ell]|}{\ell^2} = k - 2 + \frac{2(k-1)}{\ell}.\end{aligned}$$

By Corollary 3.1.9, we know

$$D(k, \mathbb{Z}^2) \leq \lim_{\ell \to \infty} \frac{1}{|T|} \left(k - 2 + \frac{2(k-1)}{\ell} \right) = 1 - \frac{2}{k}.$$

□

Theorem 3.1.10. Given a $k \in \mathbb{N}_{\geq 2}$, suppose

$$D(k, \mathbb{Z}^2) > 1 - \frac{2}{k} - \frac{1}{2k^2 - k},$$

then $D(k, (\mathbb{Z}/k\mathbb{Z})^2) = 1 - \frac{2}{k}$.

Proof. The upper bound $1 - \frac{2}{k} \geq D(k, (\mathbb{Z}/k\mathbb{Z})^2)$ is trivially true. Let $\{S_\ell\}_{\ell \in \mathbb{N}}$ be a sequence where, for each $\ell \in \mathbb{N}$, $S_\ell \in \mathfrak{G}(k, [\ell]^2)$ and $d(S_\ell) = D(k, [\ell]^2)$. Consider the frame

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$T := [0, k-1] \times [0, 2k-2]$, thanks to Corollary 3.1.9, we know the sequence of average local densities $\{\overline{d}_\ell\}_{\ell \in \mathbb{N}}$ converges to $D(k, \mathbb{Z}^2)$. By the assumption on $D(k, \mathbb{Z}^2)$, there is a $L \in \mathbb{N}$ such that the following inequality

$$\overline{d}_L > 1 - \frac{2}{k} - \frac{1}{2k^2 - k}$$

holds. For every $x \in [L]^2$, the rectangle $T + x$ can be partitioned into $2k-1$ k-lines, which implies that $d_T(x, S_L) \leq 1 - \frac{2}{k}$. Let $x_0 \in [L]^2$ be one that satisfies

$$1 - \frac{2}{k} \geq d_T(x_0, S_L) \geq \overline{d}_l > 1 - \frac{2}{k} - \frac{1}{2k^2 - k}.$$

Say $d_T(x_0, S_L) = \frac{\alpha}{2k^2 - k}$, where α is a **non-negative integer**. By above, we know

$$(2k-1)(k-2) \geq \alpha > (2k-1)(k-2) - 1,$$

which forces $d_T(x_0, S_L) = 1 - \frac{2}{k}$. For every vertical k-line l in $T + x_0$, there is a way partitioning $T + x_0$ into $2k-1$ k-lines, where l is included. It implies that every vertical or horizontal k-line has all but two positions in S_L. As a result, for every $(s, t) \in T + x_0$, if $(s, t+k) \in T + x_0$, then we have

$$(s, t) \in S_L \iff (s, t+k) \in S_L.$$

Consider the canonical projection $\pi : (T + x_0) \to (\mathbb{Z}/k\mathbb{Z})^2$. Regard $\pi(S_L)$ as a board state of $(k, (\mathbb{Z}/k\mathbb{Z})^2)$-game. Given a k-line l of torus board $(\mathbb{Z}/k\mathbb{Z})^2$. Clearly, its preimage under π contains a k-line l' in $T + x_0$. Then, by the property above, we have $\pi(l' \cap S_l) = l \cap \pi(S_L)$. As π sends each of k-lines in torus board to a k-line in plane, we get

$$|l \cap \pi(S_L)| = |\pi(l' \cap S_L)| = |l' \cap S_L| = k - 2.$$

In other words, the set $\pi(S_L)$ is a (saturated) configuration of $(k, (\mathbb{Z}/k\mathbb{Z})^2)$-game, which leads to the following lower bound

$$D(k, (\mathbb{Z}/k\mathbb{Z})^2) \geq \frac{|\pi(S_L)|}{k^2} = \frac{k(k-2)}{k^2} = 1 - \frac{2}{k}.$$

□

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Corollary 3.1.11. We have

(i) $D(3, \mathbb{Z}^2) \leq \frac{4}{15} < 1 - \frac{2}{3}$.

(ii) $D(6, \mathbb{Z}^2) \leq \frac{43}{66} < 1 - \frac{2}{6}$.

Proof. Due to the Theorem 3.1.10, it is suffice to show the toric game $(k, (\mathbb{Z}/k\mathbb{Z})^2)$ cannot reach the $1 - \frac{2}{k}$ bound when $k = 3$ or 6.

(i) For $k = 3$, it is obvious that $D(3, (\mathbb{Z}/3\mathbb{Z})^2) = \frac{1}{9}$ since a single claimed position can suppress all the other positions in this case.

(ii) For $k = 6$, the proof of $D(6, (\mathbb{Z}/6\mathbb{Z})^2) < 1 - \frac{2}{6}$ is done by the computer searching.

□

This shows that the trivial upper bound may not be tight. Let S be a saturated configuration of $(k, (\mathbb{Z}/k\mathbb{Z})^2)$-game reaching the maximum density. By tiling the S on the plane and restricting on $[\ell]^2$ for each $\ell \in \mathbb{N}$, we derive a sequence of configuration of $(k, [\ell]^2)$-game.[2] Thus we know

$$D(k, \mathbb{Z}^2) \geq \frac{|S|}{k^2} = D(k, (\mathbb{Z}/k\mathbb{Z})^2).$$

Hence, Theorem 3.1.10 implies that there is an invalid interval $(1 - \frac{2}{k} - \frac{1}{2k^2-k}, 1 - \frac{2}{k})$ for $D(k, \mathbb{Z}^2)$.

[2] Although the configurations may not be saturated, to provide a lower bound on $D(k, \mathbb{Z}^2)$, it does not matter.

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3.1.2.3 Tight Bounds



So far, for $k \in \mathbb{N}\backslash 3\mathbb{N}$, the exact value of $D(k, \mathbb{Z}^2)$ is determined. For $k \in 3\mathbb{N}$, we have the following bound

$$1 - \frac{2}{k-1} \leq D(k-1, \mathbb{Z}^2) \leq D(k, \mathbb{Z}^2) \leq 1 - \frac{2}{k},$$

which is asymptotically tight.

Theorem 3.1.12. The following equality

$$D(3, \mathbb{Z}^2) = \frac{1}{5}$$

holds.

This shows that the trivial upper bound from previous section could be tight. Also, we provide an example ($k = 3$) in which such upper bound is not tight.

For $k = 3$, we can use the following tile to fill up $[\ell]^2$.

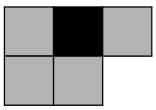


Figure 3.7: The tile: black position is claimed.

We use the tiling drawn below to tile the whole plane.

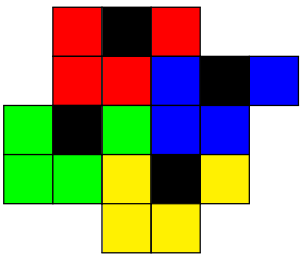


Figure 3.8: The tiling (each color represents a tile.)

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It is clear that such tiling (intersecting $[\ell]^2$) forms a saturated configuration of $(3, [\ell]^2)$-game. Call the set of claimed positions S_ℓ. Let N_ℓ be the number of tiles that is fully contained in $[\ell]^2$ and let $U_\ell \subseteq [\ell]^2$ be the union of those N_ℓ tiles. This subset U_ℓ covers all but positions within distance 2 to the border. It means that

$$[\ell]^2 \subseteq U_\ell \cup (\{1, 2, \ell-1, \ell\} \times [\ell]) \cup ([\ell] \times \{1, 2, \ell-1, \ell\}),$$

which implies

$$\begin{aligned} \ell^2 = |[\ell]^2| &\leq |U_\ell \cup (\{1, 2, \ell-1, \ell\} \times [\ell]) \cup ([\ell] \times \{1, 2, \ell-1, \ell\})| \\ &\leq |U_\ell| + |\{1, 2, \ell-1, \ell\} \times [\ell]| + |[\ell] \times \{1, 2, \ell-1, \ell\}| \\ &= 5N_\ell + 8\ell. \end{aligned}$$

In other words, we have $N_\ell \geq \frac{\ell^2 - 8\ell}{5}$. Then, by the fact that $|S_\ell| \geq N_\ell$, this configuration S_ℓ has density

$$\frac{|S_\ell|}{\ell^2} \geq \frac{N_\ell}{\ell^2} \geq \frac{\ell^2 - 8\ell}{5\ell^2}.$$

Let ℓ go to infinity, we derive

$$D(3, \mathbb{Z}^2) \geq \lim_{\ell \to \infty} D(3, [\ell]^2) \geq \lim_{\ell \to \infty} \frac{|S_\ell|}{\ell^2} \geq \frac{1}{5}.$$

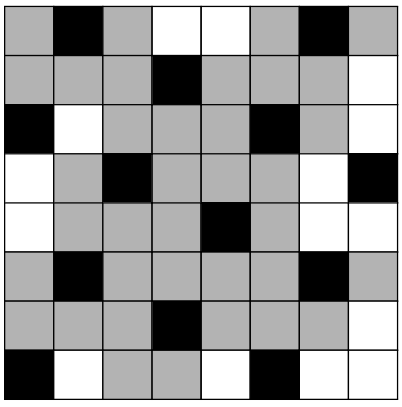


Figure 3.9: One of possible tilings for $\ell = 8$.
(White positions are uncovered by U_ℓ.)

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Now it's time to bound the local number. Let the frame T to be $[-1,1]^2$, the 3 by 3 square centered at the original. Fix a saturated configuration S of the $(3,[\ell]^2)$-game. First, if the local number of S is greater than or equal to 4 at some where, then, by pigeonhole principle, there is a row containing at least two positions, which contradicts that a configuration contains no near 3-lines. Thus we know all local numbers are bounded by 3. Second, if a frame $T+x$ contains the following structure (up to rotation and/or reflection), then each of coordinates of x equals to 2 or $\ell-1$. So this structure shows up to 4 times, which has no effect to the maximum density when we take the limit.

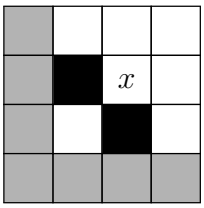


Figure 3.10: Rare structure. Shaded positions are outside $[\ell]^2$.

If $T+x$ has local number 3, then it has one position on each of rows(columns), respectively. Along with the condition on diagonals, the only possible pattern is:

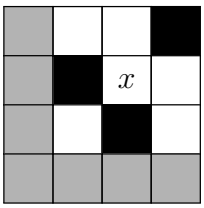


Figure 3.11: The only pattern for $n_T(x,S)=3$.

By the discussion above, we can ignore such case when we are bounding the maximum density. Let V_1 be the set of positions of local number 1, and let V_2 be the set of positions of local number 2. Clearly, we have $|V_1|+|V_2|\le \ell^2$. Build a bipartite graph H by setting

$$V(H) := V_1 \sqcup V_2,$$
$$E(H) := \{\{x,y\} : x\in V_1, y\in V_2, x-y\in T\}.$$

That is, a position x in V_1 is adjacent to all positions in the intersection of $T+x$ and V_2. It means that $\Delta(H)\le |T|-1=8$. In particular, we have $\Delta(V_1)\le 8$. On the other hand, given a position $x\in V_2$, its pattern is

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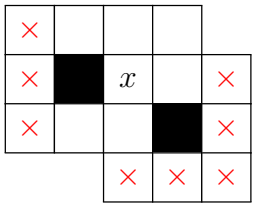


Figure 3.12: The only pattern for $n_T(x, S) = 2$.
Positions with $\times$ can not be in S.

Note that, for those two positions in $(T + x) \cap S$, their local number are 1 due to the restriction on positions with $\times$. It means $\delta(V_2) \geq 2$. Hence, we get

$$2|V_2| \leq \delta(V_2)|V_2| \leq |E(H)| \leq \Delta(V_1)|V_1| \leq 8|V_1|.$$

Then

$$\begin{aligned}\overline{n}_T([\ell]^2, S) &= \frac{|V_1| + 2|V_2|}{\ell^2} \leq \frac{|V_1| + 2|V_2|}{|V_1| + |V_2|} \\ &= 1 + \frac{|V_2|}{|V_1| + |V_2|} \leq 1 + \frac{|V_2|}{\frac{1}{4}|V_2| + |V_2|} = \frac{9}{5}.\end{aligned}$$

Finally, by Corollary 4.1.6., we derive

$$D(3, \mathbb{Z}^2) \leq \frac{9}{5|T|} = \frac{1}{5}.$$

□

Remark. We can conduct the same upper bound by using the frame $[0, 1]^2$ instead.

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3.2 Hypercube Game



3.2.1 Bounds for d: Saturated k-lines

Theorem 3.2.1. For every $k \in \mathbb{N}_{\geq 2}$ and $d \in \mathbb{N}$, we have

$$d(k, [k]^d) \leq \left(1 - \frac{2}{k}\right)^d.$$

Proof. Consider the subset $S := [2, k-1]^d$. Clearly, $S \cup \{x\}$ contains no k-line for each $x \in [k]^d \backslash S$. Define a map

$$r : [k] \to [k]$$
$$i \mapsto \begin{cases} k & \text{if } i = 1, \\ 1 & \text{if } i = k, \\ i & \text{otherwise.} \end{cases}$$

Given $x = (x_1, x_2, \cdots, x_d) \in [k]^d \backslash S$, we set $r(x) := (r(x_1), r(x_2), \cdots, r(x_d))$. First, by the choice of S, we know $x \neq r(x)$. Moreover, we can see that

$$\left\{ x + \frac{i}{k-1}(r(x) - x) : i \in [0, k-1] \right\} \subseteq S \cup \{x, r(x)\}$$

is a k-line in $[k]^d$. Hence S is a saturated configuration of $(k, [k]^d)$-game. As a consequence, the minimal density is bounded by

$$d(k, [k]^d) \leq d(S) = \frac{|S|}{k^d} = \frac{(k-2)^d}{k^d} = \left(1 - \frac{2}{k}\right)^d.$$

□

Theorem 3.2.2. For every $k \in \mathbb{N}_{\geq 2}$ and $d \in \mathbb{N}$, we have

$$d(k, [k]^d) \geq \frac{2k^d - (k+2)^d - 2d(k-2)^{d-1} + 2dk^{d-1}}{k^d}.$$

Proof. Given a saturated configuration S of $(k, [k]^d)$-game that reaches the minimal den-

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sity. By the previous theorem, we already know that

$$|S| \leq (k-2)^d.$$

Now we try to bound the number of unclaimed positions in such configuration S. Given a direction $\vec{r} \in \{-1,0,1\}^d \backslash \{(0,0,0)\}$. Say $\vec{r}$ has i non-zero coordinates. There are k^{d-i} many k-lines following this direction $\vec{r}$, and there are $\binom{d}{i}2^{i-1}$ directions having exactly i non-zero coordinates. Hence, there are at most k^{d-i} saturated k-lines following $\vec{r}$. For the case $i = 1$, the bound can be tighter. In this case, there are at most $(k-2)^{d-1}$ saturated k-lines following $\vec{r}$ since each of them covers a set of disjoint $k-2$ positions in S, respectively. Recall that, in a saturated configuration, any unclaimed position is contained in a saturated k-line. Then, as a saturated k-line contains two unclaimed positions, we get

$$\begin{aligned}
|[k]^d \backslash S| &= \#\text{ of unclaimed positions} \\
&\leq 2 \times \#\text{ of saturated } k\text{-lines for } S \\
&\leq 2\left(d(k-2)^{d-1} + \sum_{i=2}^{d} \binom{d}{i} 2^{i-1} k^{d-i}\right) \\
&= 2d(k-2)^{d-1} + (k+2)^d - k^d - 2dk^{d-1}.
\end{aligned}$$

In the other hand, we have

$$d(S) = \frac{|S|}{k^d} \geq \frac{2k^d - (k+2)^d - 2d(k-2)^{d-1} + 2dk^{d-1}}{k^d}.$$

□

To evaluate the difference between two bounds, we expand them into formal power series (in terms of k^{-1}):

$$\left(\frac{k-2}{k}\right)^d = 1 - \frac{2d}{k} + \frac{4\binom{d}{2}}{k^2} + O(k^{-3}),$$

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and

$$\begin{aligned}
&\frac{2k^d-(k+2)^d-2d(k-2)^{d-1}+2dk^{d-1}}{k^d}\\
&=\frac{1}{k^d}\left(k^d-2d(k-2)^{d-1}-\sum_{i=2}^{d}\binom{d}{i}2^ik^{d-i}\right)\\
&=1-\sum_{i=0}^{d-1}2d\binom{d-1}{i}(-2)^ik^{-i-1}-\sum_{i=2}^{d}\binom{d}{i}2^ik^{-i}\\
&=1-\frac{2d}{k}+\frac{4\binom{d}{2}}{k^2}-O(k^{-3}).
\end{aligned}$$

It shows that the difference between bounds is $O(k^{-3})$. In other words, it gives the asymptotical minimal density of $(k,[k]^d)$-game

$$d(k,[k]^d)=1-\frac{2d}{k}+\frac{4\binom{d}{2}}{k^2}\pm O(k^{-3}).$$

3.2.2 Bounds for D: Mirroring & Dimension Reduction

Fact 3.2.3. For every $k\in\mathbb{N}_{\geq 2}$ and $d\in\mathbb{N}$, we have

$$D(k,[k]^d)\leq 1-\frac{2}{k}.$$

Proof. For each $x=(x_1,x_2,\cdots,x_{d-1})\in[k]^{d-1}$, we define a k-line

$$L_x:=\{(x_1,x_2,\cdots,x_{d-1},i):i\in[k]\}.$$

The collection $\{L_x\}_{x\in[k]^{d-1}}$ of those k-lines forms a partition of $[k]^d$. Given a saturated configuration S of $(k,[k]^d)$-game, by the definition, we must have $|S\cap L_x|\leq k-2$ for each $x\in[k]^{d-1}$. It leads that

$$d(S)=\frac{|S|}{k^d}=\frac{\sum_{x\in[k]^{d-1}}|S\cap L_x|}{k^d}\leq\frac{k^{d-1}(k-2)}{k^d}=1-\frac{2}{k}.$$

As S is arbitrary chosen, we can deduce that $D(k,[k]^d)\leq 1-\frac{2}{k}$. □

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Theorem 3.2.4. For every $d \in \mathbb{N}$, there is a constant $C = C(d) > 0$ such that

$$D(k, [k]^d) \geq 1 - \frac{2}{k} - \frac{C}{k^2}$$

holds for sufficiently large $k \in \mathbb{N}$.



Proof. Let $\ell = \ell(k, d)$ be the largest integer not greater than $\frac{k}{2}$ such that it is coprime with i for every $i \in [d]$. Clearly, the difference $k - 2\ell$ is not greater than $C^* := 2\prod_{i=1}^{d} i$, a constant that only depends on d. Like the proof of Theorem 3.1.5, we define

$$Z_t(r) := \{\, \vec{x} \in \mathbb{Z}^d \;:\; \langle (1, 1, \cdots, 1), \vec{x} \rangle = r \pmod{t} \}.$$

Set $Z_d := [\ell]^d \cap Z_\ell(0)$. We claim that $|L \cap Z_d| = 1$ holds for each combinatorial line L in $[\ell]^d$. Given a combinatorial line L, there is a base point $\vec{v} \in [\ell]^d$ along with a direction $\vec{d} \in \{0, 1\}^d \backslash \{0\}$ satisfying $L = \{\vec{v} + x\vec{d} : x \in [0, \ell - 1]\}$. Given a $x \in [0, \ell - 1]$, the point $\vec{v} + x\vec{d}$ belongs to Z if and only if

$$\begin{aligned} 0 &= \langle (1, 1, \cdots, 1), \vec{v} + x\vec{d} \rangle \\ &= \|\vec{v}\|_1 + \|\vec{d}\|_1 x \pmod{\ell}. \end{aligned}$$

Note that $\|\vec{d}\|_1 \in [d]$, by the choice of ℓ, there is a unique solution to the equation. As L is arbitrary, the claim is proved.

Set $B_t := [\ell]^t \backslash Z_t$ for each $t \in \mathbb{N}$. We will use B_t as the building blocks to form a configuration S^* of $(k, [k]^d)$-game. Consider two maps: the translation by $(k - \ell)$ on $\mathbb{Z}$

$$\begin{aligned} T : \mathbb{Z} &\to \quad \mathbb{Z} \\ x &\mapsto x + (k - \ell), \end{aligned}$$

and the reflection on $[\ell]$

$$\begin{aligned} R : [\ell] &\to \quad [\ell] \\ x &\mapsto (\ell + 1) - x. \end{aligned}$$

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For every $t \in \mathbb{Z}_{\geq 0}$, the t-dimensional building block is defined to be

$$S_t := \bigcup_{\phi \in \{T,R\}^{[t]}} \phi(B_t) \subseteq [k]^t.$$

(A note is that, $[0]$ is the empty set, not $\{0\}$.)

To form S^*, we need a d-dimensional block and a number of lower dimensional blocks inserted in the gaps between the higher dimensional blocks. Given $A \subseteq [d]$ and $v \in [\ell+1, k-1]^{[d]\setminus A}$, say $d' = |A|$ and write $A = \{a_1, a_2, \cdots, a_{d'}\}$ such that $a_i < a_{i+1}$ for each $i \in [d'-1]$. Let $\phi_A : [d'] \to A$ be the bijection sending i to a_i for each $i \in [d']$. Then we derive the isomorphism

$$\begin{aligned} v\bullet : [k]^{[d']} &\xrightarrow{\sim} [k]^A \times \{v\} \subseteq [k]^{[d]} \\ x \quad &\mapsto (x \circ \phi_A^{-1}, v). \end{aligned}$$

If we regard $[k]^{[d']}$ as $[k]^{d'}$ and $[k]^A \times \{v\}$ as a subset of $[k]^d$, the image $v \bullet S_{d'}$ describes an insertion of $S_{d'}$ into the slice $[k]^A \times \{v\} \subseteq [k]^d$. For each $0 \leq t \leq d$, by collecting all possible insertions, we define

$$S_t^d := \bigcup_{A \in \binom{[d]}{t}} \bigcup_{v \in [\ell+1, k-1]^{[d]\setminus A}} v \bullet S_t \subseteq [k]^d.$$

Eventually, the configuration is defined to be

$$S^* := \bigcup_{0 \leq t \leq d} S_t^d.$$

For instant, when $k = 8$ and $d = 2$, the subset S^* looks like

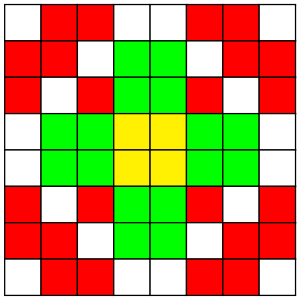


Figure 3.13: S^* for $(8, [8]^2)$-game.
(■: S_2^2, ■: S_1^2, and ■: S_0^2.)

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To show S^* is a configuration, given a k-line L. Let $\vec{v} \in \partial[k]^d$ and $\vec{d} \in \{-1, 0, 1\}^d \setminus \{0\}$ be such that $L = \{\vec{v} + i\vec{d} : i \in [0, k-1]\}$. For each $i \in [d]$, we assign it to one of three categories C_1, C_2, or C_3:

$$\begin{cases} i \in C_1 & \text{if } \vec{v}_i \in [\ell+1, k-\ell], \\ i \in C_2 & \text{if } \vec{v}_i \notin [\ell+1, k-\ell] \text{ and } \vec{d}_i = 0, \\ i \in C_3 & \text{otherwise.} \end{cases}$$

Let v be the projection of $\vec{v}$ on $[k]^{C_1}$. Let ψ be the bijection from $[d - |C_1|]$ to $C_2 \cup C_3$ preserving the ordering on $\mathbb{Z}$. Let $\phi_+, \phi_- \in \{T, R\}^{d-|C_1|}$ be two maps satisfying

$$(\phi_+)_i = \begin{cases} T & \text{if } \psi(i) \in C_2 \text{ and } \vec{v}_{\psi(i)} \in [k-\ell+1, k], \\ R & \text{if } \psi(i) \in C_2 \text{ and } \vec{v}_{\psi(i)} \in [\ell], \\ T & \text{if } \psi(i) \in C_3 \text{ and } \vec{d}_{\psi(i)} = 1, \\ R & \text{if } \psi(i) \in C_3 \text{ and } \vec{d}_{\psi(i)} = -1, \end{cases}$$

and

$$(\phi_-)_i = \begin{cases} T & \text{if } \psi(i) \in C_2 \text{ and } \vec{v}_{\psi(i)} \in [k-\ell+1, k], \\ R & \text{if } \psi(i) \in C_2 \text{ and } \vec{v}_{\psi(i)} \in [\ell], \\ R & \text{if } \psi(i) \in C_3 \text{ and } \vec{d}_{\psi(i)} = 1, \\ T & \text{if } \psi(i) \in C_3 \text{ and } \vec{d}_{\psi(i)} = -1. \end{cases}$$

Let $d' := d - |C_1|$. Now we claim that there is a combinatorial line L' of $[\ell]^{d'}$ such that both of $v \bullet \phi_+(L')$ and $v \bullet \phi_-(L')$ are contained in L. Once the claim is true, by the definition of $B_{d'}$, we know both of $v \bullet \phi_+(B_{d'})$ and $v \bullet \phi_-(B_{d'})$ misses a point from L. Note that two insertions are disjoint (since C_3 must be non-empty), the subset S^* misses at least two points from L, which implies that S^* is indeed a configuration. To prove the claim, consider the vector $\vec{u} \in [\ell]^{d'}$ and the direction $\vec{r} \in \{0, 1\}^{d'}$ satisfying

$$\vec{u}_i = \begin{cases} \vec{v}_{\psi(i)} - (k-\ell) & \text{if } \vec{v}_{\psi(i)} \in [k-\ell+1, k], \\ (\ell+1) - \vec{v}_{\psi(i)} & \text{if } \vec{v}_{\psi(i)} \in [\ell]. \end{cases}$$

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and $\vec{r}_i = -|\vec{d}_{\psi(i)}|$ for each $i \in [d']$. To show $L' := \{\vec{u}+i\vec{r} : i \in [0, \ell-1]\}$ is a combinatorial line in $[\ell]^{d'}$, we need to check that $\vec{u}_i = \ell$ when $\vec{r}_i = -1$. Notably, when $\vec{r}_i = -1$, we have $\vec{d}_{\psi(i)} = 1$ or -1. In the former, the fact $\vec{v}_{\psi(i)} = 1$ implies $\vec{u}_i = (\ell+1) - 1 = \ell$. In the latter, the fact $\vec{v}_{\psi(i)} = k$ implies $\vec{u}_i = (k) - (k-\ell) = \ell$. Given a point $x = \vec{u} + i\vec{r} \in L'$. We state the following two equalities:

$$x_+ := v \bullet \phi_+(x) = \vec{v} + (k-1-i)\vec{d}, \quad \text{and} \quad x_- := v \bullet \phi_u(x) = \vec{v} + i\vec{d}.$$

Pick an index $j \in [d]$, arbitrarily. Suppose $j \in C_1$, we have $\vec{d}_j = 0$ since $\text{supp}(\vec{d}) = C_3$ by the definition. Then, by the definition of insertion, we have

$$\begin{aligned}(x_+)_j = v_j = \vec{v}_j = (\vec{v} + (k-1-i)\vec{d})_j,\\ (x_-)_j = v_j = \vec{v}_j = (\vec{v} + i\vec{d})_j.\end{aligned}$$

Suppose $j \in C_2$, two maps $(\phi_+)_{\psi^{-1}(j)}$ and $(\phi_-)_{\psi^{-1}(j)}$ are identical. Thus

$$(x_+)_j = (x_-)_j = (\phi_+)_{\psi^{-1}(j)}(x_{\psi^{-1}(j)}) = (\phi_+)_{\psi^{-1}(j)}(\vec{u}_{\psi^{-1}(j)} + i\vec{r}_{\psi^{-1}(j)}).$$

As long as $\vec{d}_j = 0$, we know $\vec{r}_{\psi^{-1}(j)} = -|\vec{d}_{\psi(\psi^{-1}(j))}| = -|0| = 0$. Then, by the definition of $\vec{u}$, we have

$$(x_+)_j = (x_-)_j = (\phi_+)_{\psi^{-1}(j)}(\vec{u}_{\psi^{-1}(j)}) = \begin{cases} T(\vec{v}_j - (k-\ell)) = \vec{v}_j & \text{if } \vec{v}_j \in [k-\ell, k], \\ R((\ell+1) - \vec{v}_j) = \vec{v}_j & \text{if } \vec{v}_j \in [\ell]. \end{cases}$$

Recall that $\vec{d}_j = 0$ in this case, we derive

$$(x_+)_j = \vec{v}_j = (\vec{v} + (k-1-i)\vec{d})_j, \quad \text{and} \quad (x_-)_j = \vec{v}_j = (\vec{v} + i\vec{d})_j.$$

Finally, when $j \in C_3$, we have two cases: $\vec{d}_j = 1$ or $\vec{d}_j = -1$. An observation is that, we have $\vec{r}_{\psi^{-1}(j)} = -1$ and $\vec{u}_{\psi^{-1}(j)} = \ell$ in both cases.

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($= 1$) Note that, as L forms a k-line, we have $\vec{v}_j = 1$. Thus

$$\begin{aligned}(x_+)_j &= T(\ell - i) = (\ell - i) + (k - \ell) = k - i = 1 + (k - 1 - i) \\ &= (\vec{v} + (k - 1 - i)\vec{d})_j,\end{aligned}$$

and

$$(x_-)_j = R(\ell - i) = (\ell + 1) - (\ell - i) = 1 + i = (\vec{v} + i\vec{d})_j.$$

($= -1$) In this case, we have $\vec{v}_j = k$. Thus

$$\begin{aligned}(x_+)_j &= R(\ell - i) = (\ell + 1) - (\ell - i) = 1 + i = k - (k - 1 - i) \\ &= (\vec{v} + (k - 1 - i)\vec{d})_j,\end{aligned}$$

and

$$(x_-)_j = T(\ell - i) = (\ell - i) + (k - \ell) = k - i = (\vec{v} + i\vec{d})_j.$$

Then the relation between L and L' is clarified.

To count the size of S^*, we first claim that all $v \bullet S_t$ are disjoint. Given two of them, say $v_1 \bullet S_{|A_1|}$ and $v_2 \bullet S_{|A_2|}$, where $v_i \in [\ell + 1, k - \ell]^{[d]\setminus A_i}$ for $i = 1, 2$. Suppose $A_1 \neq A_2$. WLOG, we assume that $A_1 \not\supseteq A_2$. Pick and fix an index $i \in A_2 \setminus A_1$. Then, for any $x \in v_1 \bullet S_{|A_1|}$ and $y \in v_2 \bullet S_{|A_2|}$, we know

$$x_i \in [\ell + 1, k - \ell] \quad \text{and} \quad y_i \in [\ell] \cup [k - \ell + 1, k] = [k] \setminus [\ell + 1, k - \ell].$$

Suppose $A_1 = A_2$. Note $v_1 \neq v_2 \in [\ell + 1, k - \ell]^{[d]\setminus A_1}$, we can pick an index $i \in [d] \setminus A_1$ such that $(v_1)_i \neq (v_2)_i$. Then, by the definition of insertion, for any $x \in v_1 \bullet S_{|A_1|}$ and $y \in v_2 \bullet S_{|A_2|}$, we know

$$x_i = (v_1)_i \quad \text{and} \quad y_i = (v_2)_i.$$

In either cases, we have $x_i \neq y_i$ for every $x \in v_1 \bullet S_{|A_1|}$ and $y \in v_2 \bullet S_{|A_2|}$. As a result,

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the two subsets have no intersection. Thus, we have

$$|S^*| = \sum_{t=0}^{d} \sum_{S \in \binom{[d]}{t}} \sum_{v \in [\ell+1, k-\ell]^{[d]\setminus S}} |v \bullet S_t|.$$



Knowing every insertion is bijective, we get

$$|S^*| = \sum_{t=0}^{d} \sum_{S \in \binom{[d]}{t}} \sum_{v \in [\ell+1, k-\ell]^{[d]\setminus S}} |S_t| = \sum_{t=0}^{d} \binom{d}{t} (k-2\ell)^{d-t} |S_t|.$$

It remains to compute the sizes of building blocks. Similarly, we will show that $\{\phi(B_t) : \phi \in \{T, R\}^{[t]}\}$ are disjoint subsets in $[k]^t$. Given $\phi \neq \phi' \in \{T, R\}^{[t]}$. Let i be an index such that $\phi_i \neq \phi'_i$. WLOG, say $\phi_i = T$ and $\phi'_i = R$. Note that $B_t \subseteq [\ell]^t$, we have

$$\begin{aligned}(\phi(B_t))_i &= (T(B_t))_i \subseteq T([\ell]) = [k-\ell+1, k],\\ (\phi'(B_t))_i &= (R(B_t))_i \subseteq R([\ell]) = [\ell],\end{aligned}$$

i.e., $\phi(B_t) \cap \phi'(B_t) = \emptyset$. Thus we derive

$$|S_t| = |\{T, R\}^{[t]}| \times |B_t| = 2^t |B_t| = \begin{cases} 2^t \ell^{t-1}(\ell-1) & \text{if } t > 0, \\ 1 \quad (= 2^t \ell^t) & \text{if } t = 0. \end{cases}$$

Sums up, we get

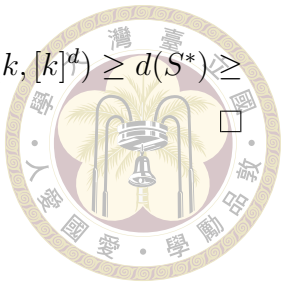
$$\begin{aligned}|S^*| &= \sum_{t=0}^{d} \binom{d}{t} (k-2\ell)^{d-t} |S_t| \\ &= \frac{\ell-1}{\ell} \sum_{t=0}^{d} \binom{d}{t} (k-2\ell)^{d-t} (2\ell)^t + \frac{1}{\ell}(k-2\ell)^d \\ &= \left(1 - \frac{1}{\ell}\right) k^d + \frac{1}{\ell}(k-2\ell)^d.\end{aligned}$$

Recall that $k - 2\ell$ is bounded by $C^* = 2 \times \prod_{i=1}^{d} i$, hence $\ell \geq \frac{1}{2}(k - C^*)$. Then the density of S^* is greater than

$$\begin{aligned}d(S^*) = \frac{|S^*|}{k^d} &= 1 - \frac{1}{\ell} + \frac{(k-2\ell)^d}{\ell k^d} \\ &= 1 - \frac{2}{k} - \frac{k-2\ell}{\ell k} + \frac{(k-2\ell)^d}{\ell k^d} \geq 1 - \frac{2}{k} - \frac{C^*}{\frac{1}{2}(k-C^*)k}.\end{aligned}$$

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As a consequence, we can find a constant $C = C(d) > 0$ such that $D(k, [k]^d) \geq d(S^*) \geq 1 - \frac{2}{k} - \frac{C}{k^2}$ for sufficiently large k. □



3.3 Winning Strategies

Since the monochromatic k in a row never draw, by some knowledge in game theory, either the first player or the second player has a winning strategy.

The following section provides some sufficient conditions that a player having a winning strategy. The result is less-generalized, but provides some incentives.

Theorem 3.3.1. Given $k, d \in \mathbb{N}$ and d positive integers $\ell_1 \leq \ell_2 \leq \cdots \leq \ell_d$ with $\ell_d \geq k$. For the $(k, \prod_{i=1}^{d}[\ell_i])$-game, we have

(i) For $k = 3$, if each ℓ_i is odd, the game is of type **FP**.

(ii) For odd k with $k > 3$, if each ℓ_i is odd and $\ell_d = k$, the game is of type **FP**.

(iii) For even k, if each ℓ_i is even and $\ell_i = k$ for some i, the game is of type **SP**.

Proof. The strategies we use for this three settings are sort of pairing strategies. We define the reflection R on the board $B = \prod_{i=1}^{d}[\ell_i]$

$$\begin{aligned} R : \quad & B \quad \to \quad B \\ & x = (x_1, x_2, \cdots, x_d) \mapsto (\ell_1 + 1 - x_1, \ell_2 + 1 - x_2, \cdots, \ell_d + 1 - x_d). \end{aligned}$$

Clearly, we have $R \circ R = \mathrm{Id}_B$. An observation is that, in the board B, L is a k-line if and only if $R(L)$ is a k-line. Then we pair up x and $R(x)$ for every $x \in B$. The only position that is not paired to other positions is the center of the board $c := (\lceil \frac{\ell_1}{2} \rceil, \lceil \frac{\ell_2}{2} \rceil, \cdots, \lceil \frac{\ell_d}{2} \rceil)$ under the setting (i) or (ii). Then we define the strategies ϕ_1, ϕ_2, and ϕ_3, separately. For convenience, we fix an arbitrary ordering σ on B. For each of settings, we first define the strategy, then explain what will it do.

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(i) The strategy ϕ_1 for the first player is defined to be

$$\phi_1 : \mathcal{F}(B) \to B \cup \{\star\}$$
$$S \mapsto \begin{cases} c & \text{if } S = \emptyset, \\ \sigma(\min X(S)) & \text{if } X(S) \neq \emptyset, \\ R(x) & \text{if } \{x\} = S \backslash R(S) \text{ and } X(S) = \emptyset, \\ \star & \text{otherwise,} \end{cases}$$

where $X(S) := \{i \in [|B|] : S \cup \{\sigma(i)\} \notin \mathcal{F}(B)\}$. First, $\phi_1(\emptyset) = c$ means that the first player would claim the center as their first move. Second, the set $\sigma(X(S))$ contains the positions that can create a k-line by adding it to the current board state S. If such position exists, by our assumption, any player should claim a position from $\sigma(X(S))$ in that moment. Thus we set $\phi_1(S) := \sigma(\min X(S))$. The third line means, after the second player claims x, the first player will claim $R(x)$ in response if there is no winning move. To check ϕ_1 is a **FP**-winning strategy, we assume the opposite. Let ϕ be a strategy such that the length of $S(\phi_1, \phi) = (s_1, s_2, \cdots, s_n)$ is even. By definition, we know $s_1 = c$ and $s_i = R(s_{i-1})$ for odd i with $1 < i < n$. Let $i < j$ be two indices such that $L := \{s_i, s_j, s_n\}$ is a 3-line. If $j \neq n-1$, then the first player can claim the position s_n before the second player do so, which leads to a contradiction. If $i \neq n-2$, then, depending on the parity of i, either s_{i-1} or s_{i+1} equals $R(s_i)$. Let i' be such index, we have $i' \leq n-2$ in either cases. Note that

$$R(L) = \{R(s_i), R(s_j), R(s_n)\} = \{s_{i'}, s_{n-2}, R(s_n)\}$$

is a 3-line and $\{s_{i'}, s_{n-2}\} \subseteq \{s_t\}_{t \in [n-2]}$. It means the first player can claim $R(s_n)$ as the $(n-1)$-th move of the game to make a first 3-line, i.e., a contradiction. Finally, if $i = n-2$, then L becomes $\{s_i, R(s_i), s_n\}$. Observe that, for every position x other than c, the only possible 3-line on the board containing both x and $R(x)$ is $\{x, R(x), c\}$. However, $s_n \neq c$ since we already have $s_1 = c$. Thus such strategy ϕ does not exist. □

(ii) We claim that ϕ_1 is a **FP**-winning strategy in this case, also. Suppose not, let ϕ be a

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strategy such that the length of $S(\phi_1, \phi) = (s_1, s_2, \cdots, s_n)$ is even. Let $i_1 < i_2 < \cdots < i_{k-1} < i_k = n$ be indices such that $\{s_{i_j}\}_{j\in[k]}$ forms a k-line. As the same reason above, we can conduct that $i_{k-1} = n-1$. If $i_{k-2} < n-2$, then the first player should claim $R(s_n)$ to create a k-line, say

$$R(L) = \{R(s_{i_j})\}_{j\in[k-1]} \cup \{R(s_n)\} \not\ni s_{i_{k-1}} = s_{n-1},$$

before the n-th move be made. If $i_{k-1} = n-2$, then L contains both of s_{n-2} and $R(s_{n-2})$. Further, due to the condition $\ell_d = k$, L is centered at c, and hence $L = R(L)$. We look at the position $\alpha := R(s_n) \in L$. If α is claimed by the second player, i.e., $\alpha = s_{2s}$ for some $s \in \mathbb{N}$ with $2s \le n-2$. Then $R(\alpha) = s_n$ should be the next claimed position s_{2s+1}, and hence $n = 2s+1 \le n-1$, a contradiction. On the other hand, if α is claimed by the first player, as $\alpha \neq c$, this move is not the first move of the game, hence we have $\alpha = s_{2s+1}$ for some $s \in \mathbb{N}$ with $2s+1 < n$. Further, note that this move does not end the game, hence it must be the reflection of the previous move. In other words, we have $s_{2s+1} = R(s_{2s})$, and hence

$$s_n = R(\alpha) = R(s_{2s+1}) = R(R(s_{2s})) = s_{2s},$$

which is impossible. □

(iii) The strategy ϕ_2 for the second player is defined to be

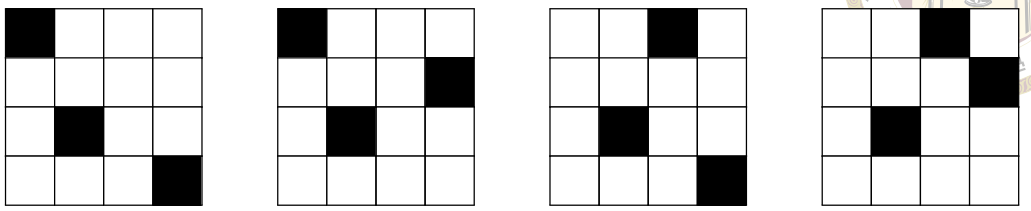
$$\phi_1 : \mathcal{F}(B) \to B \cup \{\star\}$$
$$S \mapsto \begin{cases} \sigma(\min X(S)) & \text{if } X(S) \neq \emptyset, \\ R(x) & \text{if } \{x\} = S \backslash R(S) \text{ and } X(S) = \emptyset, \\ \star & \text{otherwise,} \end{cases}$$

where $X(S)$ is what we defined above. The proof that shows ϕ_2 is a **SP**-winning strategy follows the previous proof. The only difference is that, with the condition $\ell_i = k$, $\{x, R(x)\} \subseteq L$ implies $L = R(L)$ in this case. □

Theorem 3.3.2. The $(3, [4]^2)$-game is of type **FP**.

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Proof. The following are all saturated configurations, containing the position $(2,2)$, of $(3,[4]^2)$-game.



The first position that the first player would take is $(2,2)$. By the assumption on players, the choices for the second player are $(1,4)$, $(3,4)$, $(4,1)$, and $(4,3)$. No matter what, the first player can claim a position such that those three positions claimed forms one of four configurations above, then the remaining moves are obvious. □

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Chapter 4 Extensions and Conjectures

4.1 Result from Nikodym set problem

In this section, we focus on a specific space $\mathbb{F}_q^3$, where $\mathbb{F}_q$ is the finite field of q elements. The line (q-line) structure on the space is

$$L(\mathbb{F}_q^3) = \left\{x + \mathbb{F}_q y : x \in \mathbb{F}_q^3,\ y \in (\mathbb{F}_q^3)^\times\right\}.$$

Theorem 4.1.1. Let N_s be a s-Nikodym set in $\mathbb{F}_q^3$ with q sufficiently large. If there is an injection

$$i : \mathbb{F}_q^3 \backslash N_s \to \{L \in L(\mathbb{F}_q^3) : |L \backslash N_s| = s\}$$

such that $x \in i(x)$ for every $x \in \mathbb{F}_q^3 \backslash N_s$. Then,

$$|N_s| \geq 0.38q^3.$$

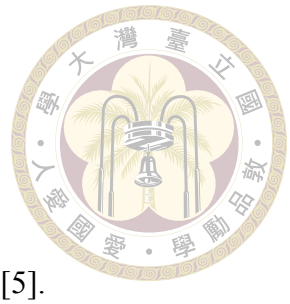
When $s = 1$, the existence of such an injection is ensured by the definition of the original Nikodym set. For higher s, the definition of s-Nikodym set only provides that there is an s-to-1 map from $\mathbb{F}_q^3 \backslash N_s$ to the set of those lines, which is not sufficient. Thus the existence of i is a non-trivial condition. In fact, this theorem is a generalised version of Theorem 1.3 in [6], where the original theorem corresponds to the case $s = 1$ in our theorem. Also, our proof follows that of Theorem 1.3 in [6].

Lemma 4.1.2 (Bipartite expander mixing lemma)**.** Let G be a biregular graph with two parts of vertices U and V. Let A be the (square) adjacency matrix of G, and let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{|U|+|V|}$ be the eigenvalues of A. Let $\lambda = \frac{\lambda_2}{\lambda_1}$. Let $S \subseteq U$ with $|S| = \alpha|U|$ and let

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$T \subseteq V$ with $|T| = \beta|V|$. Then,

$$\left| \frac{e(S,T)}{e(G)} - \alpha\beta \right| \leq \lambda\sqrt{\alpha\beta(1-\alpha)(1-\beta)}.$$



We skip the proof here, a proof of Lemma 4.1.2 can be found in [5].

Proof of Theorem 4.1.1. Consider the bipartite graph $G = (U \sqcup V, E)$ with $U := \mathbb{F}_q^3$ and $V := L(\mathbb{F}_q^3)$, where the edge set is

$$E := \{\{x, y\} : x \in U,\ y \in V,\ x \in y\}.$$

Clearly, G is a biregular graph. Moreover, we can see that

$$|U| = |\mathbb{F}_q^3| = q^3, \qquad |V| = |L(\mathbb{F}_q^3)| = \frac{q^3-1}{q-1}\frac{q^3}{q} = q^4 + q^3 + q^2 = (1+o(1))q^4,$$

and

$$e(G) = |L(\mathbb{F}_q^3) \times \mathbb{F}_q| = (1+o(1))q^5.$$

It is shown in Section 4 of [5] that

$$\lambda = \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{q^2+q}{(q^2+q+1)q}} = (1-o(1))q^{-\frac{1}{2}}.$$

Take $S := N_s$ and $T := i(\mathbb{F}_q^3 \backslash N_s)$. Then $\alpha = |N_s|q^{-3}$ and $\beta = (1-o(1))|T|q^{-4}$. By the assumption, i is injective, and hence $|N_s| = q^3 - |T|$. Note that, by the definition of T, the number of incidence between N_s and a line $L \in T$ is $|N_s \cap L| = q - s$. As a result, $e(S,T) = (q-s)|T|$. Apply Lemma 4.1.2, we get

$$\begin{aligned}
\left| \frac{(q-s)|T|}{(1+o(1))q^5} - \frac{|N_s||T|}{(1+o(1))q^7} \right| &\leq (1-o(1))q^{-\frac{1}{2}}\sqrt{(1-o(1))|N_s||T|q^{-7}(1-\alpha)(1-\beta)} \\
&\leq q^{-\frac{1}{2}}\sqrt{|N_s||T|q^{-7}(1-\alpha)(1-\beta)} \\
&= q^{-4}\sqrt{|N_s||T|(1-\alpha)(1-\beta)}.
\end{aligned}$$

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It turns

$$(q-s)|T| \le (1+o(1))\left(|N_s||T|q^{-2} + q\sqrt{|N_s||T|(1-\alpha)(1-\beta)}\right).$$

Note $|T| = q^3 - |N_s| = (1-\alpha)q^3$. Set $c := 1-\alpha$, then we have

$$\begin{aligned} cq^4 - scq^3 &= (q-s)cq^3 \\ &\le (1+o(1))\left((1-c)cq^4 + c\sqrt{(1-c)(1-(1-o(1))cq^{-1})}q^4\right) \\ &\le (1+o(1))\left((1-c)cq^4 + c\sqrt{(1-c)}q^4\right). \end{aligned}$$

By some rearrangement and simplification,

$$(1+o(1))c^2q^2 - 2(1+o(1))cqs + s^2 \le (1+o(1))(1-c)q^2.$$

As q is sufficiently large, it implies

$$c^2q^2 \le (1+o(1))(1-c)q^2,$$

i.e., $c + c^2 \le 1 + o(1)$. The solution for c is

$$\frac{-\sqrt{5}-1}{2} - o(1) \le c \le \frac{\sqrt{5}-1}{2} + o(1) \le 0.62.$$

Eventually, the size of the s-Nikodym set N_s is at least

$$|N_s| = \alpha q^3 = (1-c)q^3 \ge 0.38q^3.$$

□

4.2 Other Variants

4.2.1 Multiplayer Games

A t-player monochromatic k in a row played on the board B is denoted by (k, B, t). Each player, in turns, claims a position from the board until a k-line is created. The defi-

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nition of the game sequence remains the same. Given a game sequence $S = \{s_i\}_{i\in[n]}$, for each $i \in [t]$, the i-th player's j-th move is $s_{(j-1)t+(i-1)}$. Then here comes a natural question: Given a (k, B, t)-game, who will win the game? It seems the current two assumptions imposing on players are not sufficient to determine the winner under some games. For instance, take $(3, [3]^2, 3)$-game, there are five saturated configurations of $(3, [3]^2)$-game. There are only two types of them up to rotation and reflection.



Figure 4.1: The only two types of saturated configurations of $(3, [3]^2)$.

It means the length of any game sequence is either 3 or 5; in either cases, the first player can not win. However, the first player can "choose" which one will win. If the first move is $(2, 2)$, then the winner is the third player. If the first move is not $(2, 2)$, then the winner is the second player. Either moves are legal under our assumptions, which causes that the winner is undetermined before the first move is made. Still, we can try to determine the set of all possible winners instead of the winner. Let $\mathcal{W}(k, B, t) \subseteq [t]$ be the set of all possible winner of (k, B, t)-game.

Problem 1. Given a (k, B, t)-game, determine $\mathcal{W}(k, B, t)$.

4.2.2 Generalized configurations

We can extract the concept of saturated configuration and quantise it. Note that this type of generalised configuration does not correspond to the multiplayer game sequences.

Definition 4.2.1. Given a (k, B)-game. Let s be a positive integer. A subset $A \subseteq B$ is **s-saturated** if it satisfies

$$\forall x \in B\backslash A \; \exists L \in L(B) \text{ s.t. } x \in L \text{ and } |L\backslash A| = s,$$

and

$$\forall L \in L(B), \; |L\backslash A| \geq s.$$

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The collection of all s-saturated subsets in B is denoted by $\mathfrak{G}_s(k, B)$.



Definition 4.2.2. Given a (k, B)-game with a finite board. For every integer s with $s \geq 3$, the s**-density spectrum** of this game is defined to be the following subset

$$\mathcal{D}_s(k, B) := \{d(S) : S \in \mathfrak{G}_s(k, B)\} \subseteq [0, 1]_{\mathbb{R}}.$$

Follows it, **maximum/minimum** s**-density** of this game are

$$D_s(k, B) := \max \mathcal{D}_s(k, B) \quad \text{and} \quad d_s(k, B) := \min \mathcal{D}_s(k, B),$$

respectively.

Then the problem becomes to determine the value of $D_s(k, B)$ and $d_s(k, B)$ for any given (k, B)-game. First of all, we shall provide the existence of s-density spectrum; otherwise, the discussion becomes meaningless.

Corollary 4.2.3. Given a (k, B)-game. The only k-saturated subset is $B\backslash(\bigcup_{L\in L(B)} L)$. For s with $s > k$, there is no s-saturated subset.

Proof. Fix an integer s with $s \geq k$. Given a s-saturated subset A. For every k-line L, we have

$$k = |L| \geq |L\backslash A| \geq s \geq k.$$

It is impossible unless $s = k$. In such case, the equalities forces that $|L| = |L\backslash A|$, which means $A \cap L = \emptyset$. Since L is arbitrary, we know

$$A \cap (\cup_{L\in L(B)} L) = \emptyset.$$

In other words, we know $A \subseteq B\backslash(\cup_{L\in L(B)} L)$.

Conversely, suppose $A \subsetneq B\backslash(\cup_{L\in L(B)} L)$. Pick a x from $(B\backslash(\cup_{L\in L(B)} L))\backslash A$. Note that x is not contained in any k-lines but it belongs to $(B\backslash(\cup_{L\in L(B)} L))\backslash A \subseteq B\backslash A$, then the first condition does not hold for x, which is a contradiction. □

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It turns out that, for non-triviality, we should assume that $s < k$. In such setting, we have some similar bounds as before.



Problem 2. Given a (k, B)-game with a finite board. For every integer s with $2 < s < k$, determine the maximum and minimum s-density of this game.

Corollary 4.2.4. Given a (k, B)-game with a finite board. Let $\mathcal{F}$ be a family of disjoint k-lines, we set

$$\alpha(\mathcal{F}) := \frac{|\bigsqcup_{L\in\mathcal{F}} L|}{|B|}.$$

Then the bound

$$D_s(k, B) \leq 1 - \frac{s\alpha(\mathcal{F})}{k}$$

holds. In particular, if there is a partition of the board formed by k-lines, then $D_s(k, B) \leq 1 - \frac{s}{k}$.

Proof. Pick a s-saturated subset A reaching the maximum s-density. For every k-line $L \in \mathcal{F}$, we have $|L\backslash A| \geq s$; in other words, we have $|A\cap L| \leq k-s$. Write $X := \sqcup_{L\in\mathcal{F}} L$, then we get

$$\begin{aligned} |A| = |A \cap B| = |A \cap X| + |A \cap (B\backslash X)| &\leq \sum_{L\in\mathcal{F}} |A \cap L| + |B\backslash X| \\ &\leq \sum_{L\in\mathcal{F}} (k - s) + |B\backslash X| = \frac{|X|}{k}(k - s) + |B\backslash X| = |B| - \frac{s|X|}{k}, \end{aligned}$$

and hence

$$D_s(k, B) = d(A) = \frac{|A|}{|B|} \leq \frac{1}{|B|}\left(|B| - \frac{s|X|}{k}\right) = 1 - \frac{s\alpha(\mathcal{F})}{k}.$$

□

Corollary 4.2.5. Given a (k, B)-game with a finite board. For every $x \in B$, we define

$$L(x) := \{L \in L(B) : x \in L\}.$$

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Let $N(B) := \max_{x\in B}\{|L(x)|\}$. Then we have

$$d_s(k,B) \geq \frac{k-s}{sN(B)+(k-s)}.$$



Proof. The proof mimics the proof of Theorem 3.1.1. Let S be a s-saturated subset. Set $N := B\backslash A$. Consider the bipartite graph $G = (N \sqcup S, E)$ defined by

$$E := \{\{x,y\} : x \in N,\ y \in S,\ \text{there is a } k\text{-line } L \ni x,y \text{ s.t. } |L\backslash S| = s\}.$$

Given a position $x \in N$, by the definition of s-saturated subset, there is a k-line $L \in L(B)$ with $x \in L$ such that $|L\backslash S| = s$. Thus we have $L \cap S \subseteq N_G(x)$. As x is arbitrary, we get $\delta(N) \geq |L \cap S| = k - s$. On the other hand, given a position $y \in S$, we double count the pair (L, x) where $L \in L(y)$ and $x \in N_G(y)$ satisfying $x \in L$ and $|L\backslash S| = s$. First, by the definition of the graph G, we have

$$|N_G(y)| = \sum_{x\in N_G(y)} 1 \leq \sum_{x\in N_G(y)} \left(\sum_{L\in L(x)\cap L(y):|L\backslash S|=s} 1 \right) = \#\text{ of pairs.}$$

Count the k-line first, we get

$$\begin{aligned} \#\text{ of pairs} &= \sum_{L\in L(y):|L\backslash S|=s} \left(\sum_{x\in L\cap N} 1 \right) = \sum_{L\in L(y):|L\backslash S|=s} |L\backslash S| \\ &= \sum_{L\in L(y):|L\backslash S|=s} s \leq s|L(y)| \leq sN. \end{aligned}$$

As this inequality holds for every $y \in S$, we derive $\Delta(S) \leq sN(B)$. Then we have

$$(k-s)|N| \leq \delta(N)|N| \leq |E| \leq \Delta(S)|S| \leq sN(B)|S|.$$

Adding $(k-s)|S|$ to both sides, we get

$$(k-s)|B| = (k-s)(|N|+|S|) \leq (sN(B)+(k-s))|S|,$$

which means

$$d(S) = \frac{|S|}{|B|} \geq \frac{k-s}{sN(B)+(k-s)}.$$

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As S is arbitrary, we may conclude that $d_s(k, B) \geq \frac{k-s}{sN+(k-s)}$. □



Remark. This bound is weaker than Theorem 3.1.1. Directly apply this corollary to $(k, [\ell]^2)$-game, as we know $N([\ell]^2) = 4k$ for $\ell \geq 2k-1$, we get

$$d(k, [\ell]^2) = d_2(k, [\ell]^2) \geq \frac{k-2}{8k+(k-2)} \sim \frac{1}{9}$$

for $\ell \geq 2k-1$. In contrast, Theorem 3.1.1 provides $d(k, [\ell]^2) \gtrsim 1 - \frac{16}{k}$.

4.2.3 Board-Shape Problem

The original monochromatic k in a row problem asks: "how high (or low) the density of a saturated configuration of a given board can be?" We can ask a similar problem: "For a fixed board size, what shape will maximize (or minimize) the range of density spectrum?" Still, this problem is too arbitrary. Thus we will stick on a well-structured ground board, which is the plane board $\mathbb{Z}^2$ along with the k-line structure

$$L(\mathbb{Z}^2) := \{x + [0, k-1]y : x, y \in \mathbb{Z}^2\}.$$

For every subset $B \subseteq \mathbb{Z}^2$, we regard it as a board by setting

$$L(B) := \{L \in L(\mathbb{Z}^2) : L \subseteq B\}$$

to be its k-line structure. Then, for a given size $n \in \mathbb{N}$, the family $\binom{\mathbb{Z}^2}{n}$ of boards is what we concerned.

Definition 4.2.6. Given $k, n \in \mathbb{N}$, we define the extreme values

$$D_{min}(k, n) := \min_{B \in \binom{\mathbb{Z}^2}{n}} D(k, B),$$

and

$$d_{max}(k, n) := \max_{B \in \binom{\mathbb{Z}^2}{n}} d(k, B).$$

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Lemma 4.2.7. Given a (k, B)-game with a finite board. We have

$$\bigcap_{A\in\mathfrak{G}(k,B)} A \supseteq B\backslash(\cup_{L\in L(B)}L),$$

which means that every saturated configuration A covers the subset $B\backslash(\cup_{L\in L(B)}L)$.

Proof. Suppose otherwise, there is a saturated configuration $A \subseteq B$ and a position $x \in B\backslash(\cup_{L\in L(B)}L)$ such that $x \notin A$. By the property of saturated configurations, there is a k-line $L_0 \in L(B)$ such that $x \in L_0$ and $|L_0\backslash A| = 2$. Hence, we get

$$x \in L_0 \cap (B\backslash(\cup_{L\in L(B)}L)) \subseteq L_0 \cap (B\backslash L_0) = \emptyset,$$

a contradiction. □

Definition 4.2.8. Given a (k, B)-game with a finite board. The set $B\backslash(\cup_{L\in L(B)}L)$ is called the **kernel**, or the **core**, of the board B, and is denoted by $\ker(B)$.

Example 4.2.9. For every $n \in \mathbb{N}$, consider the subset

$$B_n := \bigcup_{s\in[n]} ([1] \times \{2s + 1\}) \subseteq \mathbb{Z}^2.$$

Clearly, B_n contains no k-line, hence $\ker(B_n) = B_n$. Then, by Lemma 4.2.7, we have

$$d(k, B_n) \geq \frac{|\ker(B_n)|}{|B_n|} = \frac{|B_n|}{|B_n|} = 1.$$

It implies that $D(k, B_n) = d(k, B_n) = 1$. Therefore, as $|B_n| = n$, we get

$$d_{max}(k, n) = 1.$$

It shows that the problem becomes less interesting once the core is huge in the scale of n. Thus we shall restrict the problem on families of core-less subsets.

Definition 4.2.10. For every $n \in \mathbb{N}$, we define

$$\binom{\mathbb{Z}^2}{n}^{\times} := \left\{B \in \binom{\mathbb{Z}^2}{n} : \ker(B) = \emptyset\right\} \subseteq \binom{\mathbb{Z}^2}{n}$$

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to be the family of core-less n-subsets in $\mathbb{Z}^2$. Then the adjusted extreme values are

$$D^{\times}_{min}(k,n) := \min_{B\in\binom{\mathbb{Z}^2}{n}^{\times}} D(k,B),$$

and

$$d^{\times}_{max}(k,n) := \max_{B\in\binom{\mathbb{Z}^2}{n}^{\times}} d(k,B).$$

Besides, there are two families of boards

$$\mathcal{F}_D(k,n) := \{B \in \binom{\mathbb{Z}^2}{n}^{\times} : D(k,B) = D^{\times}_{min}(k,n)\},$$

and

$$\mathcal{F}_d(k,n) := \{B \in \binom{\mathbb{Z}^2}{n}^{\times} : d(k,B) = d^{\times}_{max}(k,n)\},$$

that we are concerned.

Problem 3. Given $k \in \mathbb{N}$, determine the behavior of

$$D^{\times}_{min}(k,n), \text{ and } d^{\times}_{max}(k,n)$$

as n goes to infinity. For every $n \in \mathbb{N}$, characterise $\mathcal{F}_D(k,n)$ and $\mathcal{F}_d(k,n)$.

4.3 Conjectures

From Theorem 3.1.6 and the proof of Corollary 3.1.11, the inequalities

$$D(k,(\mathbb{Z}/k\mathbb{Z})^2) \le D(k,\mathbb{Z}^2) \le 1 - \frac{2}{k}$$

hold for every $k \in \mathbb{Z}_{\ge 2}$. Note that, from the construction of Theorem 3.1.5, both inequalities are equal when k is not a multiple of 3. Further, the first inequality is strict when $k = 3$, and the second inequality is strict when $k = 3$ or 6, which inspire us the following conjectures.

Conjecture 4.3.1. For every $k \in 3\mathbb{N}$, $D(k,\mathbb{Z}^2) < 1 - \frac{2}{k}$.

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Conjecture 4.3.2. For every $k \in 3\mathbb{N}$, $D(k, (\mathbb{Z}/k\mathbb{Z})^2) < D(k, \mathbb{Z}^2)$.

Suppose the Conjecture 4.3.1 is true, then the inequality

$$D(k, \mathbb{Z}^2) \geq D(k, (\mathbb{Z}/k\mathbb{Z})^2)$$

forces the bound $D(k, (\mathbb{Z}/k\mathbb{Z})^2) < 1 - \frac{2}{k}$ to hold for $k \in 3\mathbb{N}$. Conversely, according to Theorem 3.1.10, if $D(k, (\mathbb{Z}/k\mathbb{Z})^2) < 1 - \frac{2}{k}$ holds for some k, then we have

$$D(k, \mathbb{Z}^2) \leq 1 - \frac{2}{k} - \frac{1}{2k^2 - k} < 1 - \frac{2}{k}$$

for the same k. Hence, that conjecture is equivalent to

$$\left[\forall k \in 3\mathbb{N}, D(k, (\mathbb{Z}/k\mathbb{Z}))^2) < 1 - \frac{2}{k}\right].$$

In fact, this conjecture can be stated in a more general way, without using our density notation. Given a subset $A \subseteq (\mathbb{Z}/k\mathbb{Z})^2$ and let $S := (\mathbb{Z}/k\mathbb{Z})^2 \backslash A$. Recall that if S is 2-saturated, then it is a saturated configuration of $(k, (\mathbb{Z}/k\mathbb{Z})^2)$-game, and vise versa. Therefore, the conjecture asks if there is a S with $|S| = k^2 - 2k$; equivalently, it asks if there is a A with $|A| = 2k$, which is the following question.

Question. Is there a subset $A \subseteq (\mathbb{Z}/k\mathbb{Z})^2$ satisfying $|L \cap A| = 2$ for every line L in $(\mathbb{Z}/k\mathbb{Z})^2$?

For the minimum density, recall that we have $1 - \frac{16}{k} \lesssim d(k, \mathbb{Z}^2) \lesssim 1 - \frac{8}{k}$ as k goes infinity. What is the correct constant over k? The only thing we can tell is that it would not be 16. If it does, then the average degree of two parts of G defined in Theorem 3.1.1 should be quite close to $k - 2$ and 16, respectively. It means that, in the graph G, the average degree on one side should be minimized while that on the other side should be maximized, which seems impossible to fulfill.

Question 4.3.3. Determine the correct constant $c \in [8, 16]$ such that

$$d(k, \mathbb{Z}^2) = 1 - \frac{c}{k} + o(k^{-1}).$$

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Finally, for winning strategies, according to Theorem 3.3.1, $(3, [\ell]^2)$-game is of type **FP** if ℓ is odd. Also, by brute force, we can show that both of $(3, [4]^2)$-game and $(3, [6]^2)$-game are of type **FP**. Then we boldly conjecture that this phenomenon holds on every square board of even side length.

Conjecture 4.3.4. For every $\ell \in 2\mathbb{N}$ with $\ell \geq 8$, the $(3, [\ell]^2)$-game is of type **FP**.

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